

Deterministic Polynomial Factorization over Prime Fields

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Abstract

We give a uniform deterministic polynomial-time algorithm for complete factorization over prime fields. For a prime p in binary and a nonzero polynomial $f \in \mathbf{F}_p[x]$ given by its dense coefficient list, the algorithm computes the irreducible factors and their multiplicities using a number of bit operations polynomial in $(\deg f + 1) \log p$. The proof uses the uniform Hecke zero-free theorem from the companion paper *Primitive roots for every admissible integer base*.

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1 Introduction

Let p be a prime, given in binary, and let $f \in \mathbf{F}_p[x]$ be a nonzero polynomial, given by its dense coefficient list. Write $n = \deg f$ and $L = \lceil \log_2 p \rceil$. Complete factorization means computing the leading coefficient c and distinct monic irreducible polynomials g_i , with positive integer multiplicities e_i , such that

$$f = c \prod_i g_i^{e_i}.$$

The empty product is permitted when $n = 0$. The complexity parameter is $(n + 1)L$, so an algorithm polynomial in p does not give a polynomial-time algorithm in the prescribed representation.

Finite-field factorization is a basic constructive problem in algebra. It also exposes a persistent distinction between randomized and deterministic computation: a random element of a product of finite fields often separates its components, whereas constructing a separator uniformly can require additional arithmetic information. The present paper isolates one form of that information and gives a deterministic splitting procedure once it is available. To supply it uniformly, we use the zero-free theorem for finite-order Hecke L -functions in the companion paper [21, Theorem 1.2]. We state that input precisely in Theorem 11.1 and prove its auxiliary-prime consequence here. The analytic theorem itself belongs to the companion.

Theorem 1.1. *There is a uniform deterministic algorithm that, given a prime p in binary and a nonzero dense polynomial $f \in \mathbf{F}_p[x]$ of degree n , outputs its leading coefficient and its distinct monic irreducible factors with their multiplicities in*

$$O(((n + 1)\lceil \log_2 p \rceil)^{10^{12}})$$

bit operations, with absolute implied constant. A constant polynomial returns an empty factor list. The algorithm uses no randomness, integer-factorization oracle, primitive-root oracle, or GRH assumption.

The exponent is deliberately generous. The point is one fixed polynomial bound for all degrees and prime characteristics. We separate the algebraic construction from its analytic input in the following reduction.

1.1 The auxiliary-prime reduction

Put

$$B = 20 + (n + 1)(L + 1). \quad (1.1)$$

For a prime $q \leq n$, call a rational prime ℓ an *auxiliary prime* for (p, q) if

$$\ell \notin \{2, 3, p, q\}, \quad \ell \equiv 1 \pmod{12q}, \quad p^{(\ell-1)/q} \not\equiv 1 \pmod{\ell}. \quad (1.2)$$

Thus p is not a q th power modulo ℓ . The congruence modulo $12q$ is convenient for the analytic construction in Section 11; the algebraic construction only requires the corresponding congruence modulo q .

Theorem 1.2. *There is a uniform deterministic algorithm with the following guarantee. Its input is a prime p and a nonzero dense polynomial $f \in \mathbf{F}_p[x]$. If $p > B^{200000}$ and $n \geq 2$, its input also includes one auxiliary prime ℓ_q satisfying (1.2) for each prime $q \leq n$. It outputs the complete factorization of f , including multiplicities. Let $E = \max(2, \max_q \ell_q)$, with $E = 2$ when no auxiliary primes are required. For an absolute constant C , the algorithm uses $O((B + E)^C)$ bit operations. It uses neither randomness nor an integer factorization or primitive-root oracle.*

The dependence on E is on the numerical value of the auxiliary primes, not merely their bit lengths. Consequently Theorem 1.2 becomes a polynomial-time factorization theorem when these primes are bounded by a fixed power of B . Upward search, trial division, and modular exponentiation then construct them within the same kind of bound. We prove this consequence from the cited zero-free theorem in Section 11. The reduction in Theorem 1.2 does not use that theorem.

1.2 History and the role of the construction

Berlekamp's algorithms put linear algebra at the center of finite-field factorization: Frobenius-fixed elements in the quotient algebra encode its irreducible factors, and polynomial greatest common divisors turn suitable such elements into factors [2, 3]. Randomized algorithms, including Cantor–Zassenhaus [4], obtain expected polynomial running time by finding separating elements with random choices. The difficulty studied here is to make every such choice deterministically, with a bound polynomial in both the degree and the bit length of the characteristic.

Several deterministic results delineate this difficulty. Shoup's algorithm has polynomial dependence on the degree and a $p^{1/2+o(1)}$ dependence on the characteristic [28]. Under the generalized Riemann hypothesis, Rónyai obtained polynomial bounds when the number of irreducible factors is bounded [25], and Evdokimov obtained a bound $(n^{\log n} \log p)^{O(1)}$ for general degree [7]. The latter is quasipolynomial in n . The reduction proved here, combined with the fixed-width zero-free strip that follows immediately from Hecke GRH, already gives a polynomial bound in both parameters under GRH. For Theorem 1.1, the companion supplies that strip without GRH. In particular, the hypotheses and parameters

of a deterministic result matter: removing random choices alone does not establish a uniform polynomial bound in $(n + 1)L$.

Geometric methods already have an important place in this history. Schoof’s elliptic-curve method computes a square root of a fixed integer a modulo a varying prime, when one exists, in time polynomial in $\log p$; the uniform bound in the two parameters is polynomial in $|a|$ and $\log p$, rather than in their binary lengths [27, Section 4]. Pila’s computation of Frobenius on abelian varieties yields deterministic construction of roots of unity of fixed odd prime order modulo primes congruent to 1 modulo that order, with a running-time exponent that may depend on the order and on the geometric presentation [22, Theorems A and D]. These results motivate careful accounting for the varying curve and prime in the present construction. We must bound the degrees of field extensions, the sizes of divisor representations, and the dimensions of all linear systems by fixed powers of the original input parameters.

For a single prime not dividing the discriminant of an integral lift, Rónyai obtained, under GRH, deterministic bounds involving the degree of that lift’s splitting field [26]. A different kind of uniformity arises when reductions modulo many primes are processed together. Altman’s unconditional algorithm takes a monic irreducible polynomial in $\mathbb{Z}[x]$ of coefficient height at most $H \geq 2$, with splitting-field degree m , and factors its reductions modulo every prime below X in total time

$$\pi(X)(m \log H)^{O(1)} \log^5 X.$$

This is an amortized result over primes [1, Theorem 1.1]. The parameter m can be as large as the factorial of the polynomial’s degree. Thus this result does not give the worst-case single-input bound sought here. Altman’s introduction also provides a recent account of the general deterministic problem and its remaining obstacles.

The algebraic techniques used below have substantial precedents. Computation at unknown roots by working in a quotient algebra and splitting it when a zero test distinguishes components is an instance of dynamic evaluation [5, 6]. The digit calculations in primary multiplicative groups are the prime-power part of the Pohlig–Hellman algorithm [23]. Rónyai already proved that a completely split polynomial of even degree can be split in polynomial time when a quadratic nonresidue is supplied [25]; we include an explicit tournament proof suited to our representation. The odd-degree construction uses cyclic curves $Y^q = F(X)$. The cyclotomic action on their Jacobians and descriptions of its first torsion layer by ramification divisors are familiar from explicit descent, notably the work of Poonen and Schaefer [24, Sections 4 and 6].

Our effective treatment of divisors also builds on established computational geometry. Ideal representations on superelliptic curves, Riemann–Roch algorithms, and finite-dimensional linear algebra for Jacobian arithmetic appear in [8, 13, 16]. For norm equations, we use the relation between cyclic algebras and norms, and the approach through maximal orders and minimal left ideals developed in explicit matrix-algebra computations [9, 15, 14]. The general function-field algorithm of Ivanyos, Kutas, and Rónyai uses a finite-field polynomial-factorization oracle. Here the cyclic presentation gives explicit local orders, and evaluation of their global sections at a rational point gives a rank-one idempotent by linear algebra. The structural explanation uses the splitting of vector bundles on the projective line [10, 12].

The substantive step of the present reduction is the odd-degree separator, together with a uniform implementation of the divisor-class divisions it requires. A filtration of lifted ramification classes keeps the necessary field extensions of polynomial degree. A lattice of divisors at infinity then forces two ramification points to receive different labels. We prove this mechanism and its algorithms directly; the earlier results just discussed provide their

mathematical context. The separate step of finding sufficiently small auxiliary primes is isolated in Section 11, which is the only place the companion’s analytic theorem enters.

1.3 Proof strategy

The Berlekamp algebra reduces complete factorization to splitting a monic square-free polynomial F all of whose roots already belong to $\mathbf{F}_p$. Write $N = \deg F \geq 2$. The unknown roots are represented simultaneously by the finite algebra $\mathbf{F}_p[T]/(F)$; an operation whose outcomes differ between components immediately produces a factor by a greatest common divisor. We can therefore describe computations at an unspecified root and carry them out without first finding that root.

The auxiliary primes construct the multiplicative information needed in these computations. For each odd prime q , they give a field $K_q = \mathbf{F}_p(\zeta_q)$ and a generator of the full q -primary subgroup of K_q^* . The entry for $q = 2$ is a generator of the full 2-primary subgroup of $\mathbf{F}_p^*$. Here the q -primary subgroup consists of the elements whose orders are powers of q . These entries permit extraction of promised q th roots in finite algebras by small-prime discrete logarithms. Their construction proceeds in increasing q : the only factorization needed to make the entry for q has degree $q - 1$.

When N is even, differences between pairs of roots and the 2-primary entry orient the complete graph on the roots. Opposite differences have opposite orientations. The vertex scores cannot all be equal when N is even, so they provide a separator. The more substantial construction handles odd N .

Choose an odd prime $q \mid N$, and write $r = v_q(N)$. Over an explicitly constructed field k of degree at most $N(q - 1)$ over $\mathbf{F}_p$, consider the smooth projective curve with affine equation

$$C : Y^q = F(X).$$

Its ramification points are $P_i = (x_i, 0)$, where the x_i are the unknown roots of F . There are q rational points at infinity. The automorphism $\sigma(X, Y) = (X, \zeta_q Y)$ acts on degree-zero divisor classes; write $\lambda = 1 - \sigma$. Degree-zero divisors supported at infinity form a lattice on which λ is injective. In contrast, λ is surjective on the geometric Jacobian.

This difference is the source of the splitting argument. We lift each class $[P_i - \infty_0]$ repeatedly through λ , keeping track of an infinity divisor at every stage. A filtration argument shows that all the classes required for these lifts are rational over the field k ; the extension degree stays polynomial even though the number of possible classes is large. Summing the lifted classes and applying explicit label tests forces a distinction between ramification points. If every test gave one common label, a fixed nonzero infinity divisor would become divisible by one more power of λ than its lattice valuation allows.

The geometric existence argument is accompanied by a constructive division algorithm. A promised λ -division reduces to a norm equation in $k(C)/k(X)$. We solve that equation by constructing maximal orders in a split cyclic algebra and computing their global sections on the projective line. A rank-one idempotent in this algebra yields the norm solution. All computations use polynomial and matrix arithmetic; Riemann–Roch reduction controls divisor sizes, and arithmetic circuits store principal functions arising from large infinity multiplicities. Figure 1 records the roles of these ingredients.

The technical results are uniform in the degree of the curve and the prime q . This uniformity is essential: a running-time exponent depending on the genus would not suffice for Theorem 1.2. We give the finite-field tools first, then the auxiliary table, the geometry, and the effective divisor and norm calculations. The splitter and the factorization driver assemble these ingredients. The final two sections prove the bit bounds and the analytic implication.

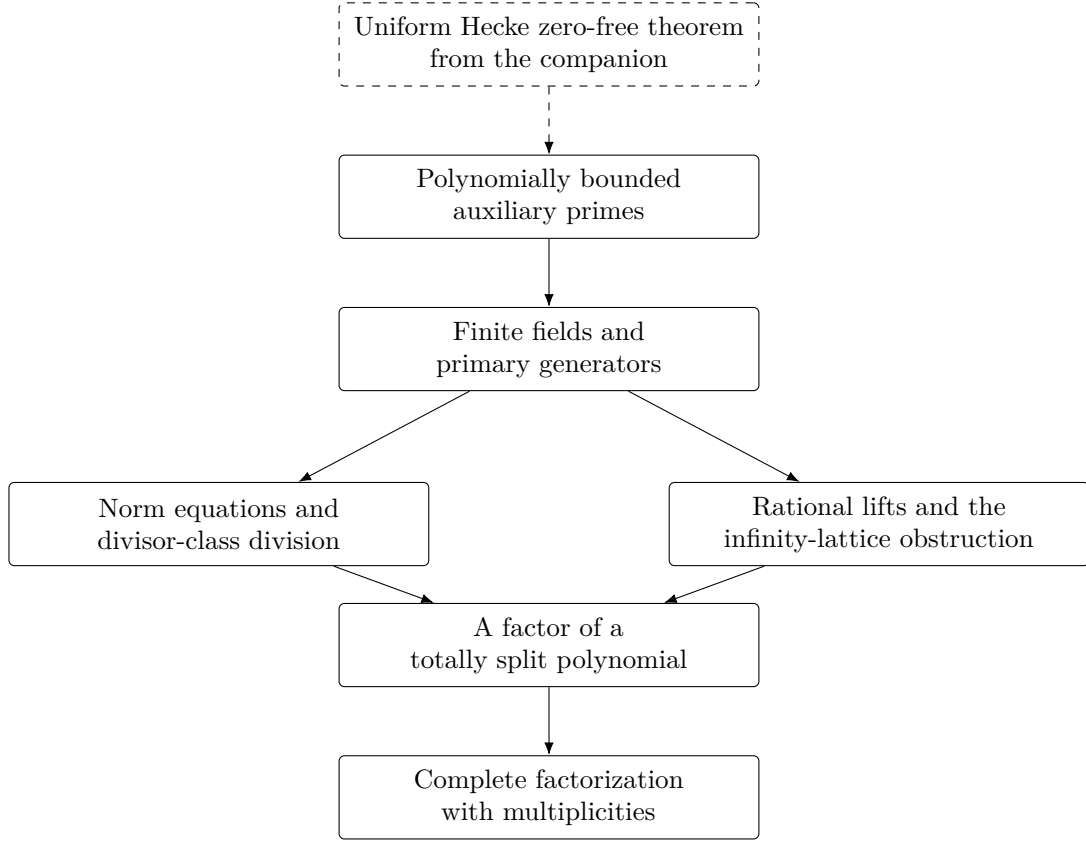


Figure 1: The proof separates auxiliary-prime existence from the algebraic reduction. The dashed implication uses the cited analytic theorem; the remaining constructions are proved in the paper. The two middle branches are the effective and geometric parts of the odd-degree splitter.

2 Finite-field computations without known roots

The splitting procedures will manipulate all roots of a polynomial at once. Two elementary tools make this possible. First, a computation over a field can be executed in a product of fields until a zero test distinguishes two components. Second, a supplied generator of a primary multiplicative subgroup permits logarithms and root extraction by short digit calculations. We develop both tools explicitly, and then use them to split every totally split polynomial of even degree.

Finite fields are represented by a basis over their prime field and a multiplication table, or by an equivalent tower of polynomial quotients. Field arithmetic, binary powering, polynomial Euclidean algorithms, and linear algebra are therefore deterministic procedures with polynomial bit complexity in the representation lengths. All basis choices and pivot choices below use a fixed ordering. For a prime q and a positive integer m , write $v_q(m)$ for its q -adic valuation. If k has cardinality Q and $q \mid Q - 1$, its full q -primary subgroup is the unique subgroup of k^* of order $q^{v_q(Q-1)}$.

2.1 Simultaneous execution and zero tests

Let k be an explicitly represented finite extension of $\mathbf{F}_p$, and let $F \in \mathbf{F}_p[T]$ be monic, square-free, and totally split, of degree N . Write its roots as $x_1, \dots, x_N$ for the proof; the

algorithm does not know these roots. The algebra

$$A = k[T]/(F) \simeq \prod_{i=1}^N k, \quad a(T) \longmapsto (a(x_1), \dots, a(x_N)) \quad (2.1)$$

lets us carry out arithmetic on all components simultaneously.

Lemma 2.1 (Simultaneous execution). *Suppose a deterministic field procedure uses arithmetic, zero tests, and integer control, and its scalar input at component i is obtained by specializing given elements of A at $T = x_i$. Assume the procedure is defined for each specialized input. One can either execute the procedure on all components with a common sequence of decisions, or return a nontrivial proper monic divisor of F in $\mathbf{F}_p[T]$. The extra cost of each scalar operation or zero test is polynomial in N , $[k : \mathbf{F}_p]$, and $\log p$.*

Proof. For a zero test on $a \in A$, compute $d = \gcd(F, a)$ over k using its representative of degree less than N . If $d = 1$, every component is nonzero; if $d = F$, every component is zero. In the remaining case, d is a proper factor and the procedure stops. Although computed over k , its monic form belongs to $\mathbf{F}_p[T]$: it is the product of $T - x_i$ over the components where $a(x_i) = 0$.

Addition and multiplication take place in A . A nonzero scalar is inverted only after the zero test has shown it nonzero in every component; the extended Euclidean algorithm then gives its inverse modulo F . Thus, until a split occurs, every field instruction and decision is valid in every component. Polynomial degree tests and choices of pivots reduce to successive zero tests on coefficients, so the same argument applies inside polynomial arithmetic and linear algebra. The usual dense implementations give the cost bound. $\square$

This is the dynamic-evaluation principle [5]; we call its use here *simultaneous execution*. In particular, a divergent pivot or a divergent polynomial degree is useful information: it already supplies the required factor. The convention also applies to nested procedures whose coefficient field is being simulated by A .

2.2 Logarithms in primary subgroups

Let q be a prime, and suppose u generates a cyclic group of order q^s . For $a \in \langle u \rangle$, its logarithm $\ell \in [0, q^s)$ can be recovered one base- q digit at a time, as in the prime-power logarithm algorithm of Pohlig and Hellman [23]. If $\ell_j \equiv \ell \pmod{q^j}$ is already known, the next digit is the unique $c \in \{0, \dots, q-1\}$ satisfying

$$(au^{-\ell_j})^{q^{s-j-1}} = (u^{q^{s-1}})^c, \quad 0 \leq j < s. \quad (2.2)$$

Set $\ell_{j+1} = \ell_j + cq^j$ and begin with $\ell_0 = 0$. Indeed, if $\ell - \ell_j = q^j(c + qv)$, raising to q^{s-j-1} removes the term qv and leaves exactly Equation (2.2). This uses q equality tests per digit, rather than an enumeration of q^s group elements.

The same calculation works in a product of fields when u has order q^s in every component. When the product is presented as $k[X]/H$ with H square-free, gcds with H partition the components at each digit according to the equality in Equation (2.2); retain each nonempty piece and its current digit string. There are at most $\deg H$ pieces at every level. Keep each final integer logarithm together with its polynomial modulus. Algebra elements subsequently computed from these logarithms can be recombined by polynomial Chinese remaindering. Alternatively, when zero components are represented by idempotent masks, perform the same partitions using those masks. These two implementations will be used below.

2.3 Root extraction in a finite reduced algebra

We next give the root procedure used in the norm equations later in the paper. Its input already includes a generator over the constant field, but it does not include generators in extension fields, nor a factorization of the modulus.

Lemma 2.2 (Root extraction from a primary generator). *Let k be an explicit finite field of cardinality Q and characteristic p . Let q be an odd prime dividing $Q - 1$, let $\zeta \in k$ have order q , and let $\omega \in k$ generate the full q -primary subgroup of k^* . Let $H \in k[X]$ be monic and square-free, with $h = \deg H \geq 1$, and suppose $p > h^2$. Given a unit $a \in k[X]/H$ that is promised to be a q -th power, one can deterministically construct b with $b^q = a$. The bit cost is polynomial in q , h , and the length of the given field representation. The procedure uses only arithmetic and zero tests over k , and so admits simultaneous execution as in Lemma 2.1.*

Proof. The construction has three steps: separate extension degrees, obtain a primary generator on each resulting part, and take the two coprime parts of the root.

Separate extension degrees. For $d = 1, \dots, h$, use gcds with $X^{Q^d} - X$, removing factors already found, to form the product H_d of the irreducible factors of H of degree d . All powers are computed modulo the current modulus by binary powering. Discard $H_d = 1$. Every field component of $k[X]/H_d$ has cardinality Q^d . Write

$$Q^d - 1 = q^s t, \quad q \nmid t.$$

Only division by the specified prime q is needed to find s and t .

Obtain a primary generator. Since q is odd and $q \mid Q - 1$, the elementary lifting-the-exponent identity is

$$v_q(Q^d - 1) = v_q(Q - 1) + v_q(d). \quad (2.3)$$

To see this, raising a number congruent to 1 modulo q to an exponent prime to q preserves the valuation of its difference from 1, whereas raising it to the q -th power increases that valuation by exactly one. Both assertions follow from the binomial expansion: its linear term has strictly smaller q -valuation than all subsequent terms, using that q is odd in the second assertion. Iterating proves Equation (2.3). If $q \nmid d$, set $u = \omega$, which already has order q^s in every component. Suppose instead that $q \mid d$. Solve the k -linear equation

$$v^{Q^{d/q}} = \zeta v \quad \text{in } k[X]/H_d. \quad (2.4)$$

In each component its nonzero solutions exist: the cyclic group of order $Q^d - 1$ contains a solution of $v^{Q^{d/q}-1} = \zeta$, because

$$\zeta^{(Q^d-1)/(Q^{d/q}-1)} = 1.$$

For such a solution, Equation (2.3) shows that the q -valuation of its order is s .

To choose a solution nonzero in every component without knowing those components, take a kernel basis $v_0, \dots, v_{r-1}$ and try

$$v(c) = \sum_{j=0}^{r-1} c^j v_j, \quad c = 0, \dots, h^2. \quad (2.5)$$

Test each candidate for being a unit by a gcd with H_d . In any component, at least one basis vector is nonzero, so at most $r - 1$ values of c make $v(c)$ zero there. There are at most h components and $r \leq h$; fewer than h^2 values are excluded altogether. The hypothesis

$p > h^2$ makes the listed parameters distinct. Hence the search succeeds. Raising its result to t produces a generator u of order q^s in every component.

Take the root. Use the integer Chinese remainder theorem for q^s and t to choose exponents projecting a to its primary and prime-to- q parts. Explicitly, put

$$a_q = a^{t(t^{-1} \bmod q^s)}, \quad a_t = a^{q^s((q^s)^{-1} \bmod t)},$$

with $a_t = 1$ when $t = 1$. Then $a = a_q a_t$, the order of a_q divides q^s , and the order of a_t divides t . A root of a_t is obtained by raising it to $q^{-1} \bmod t$, again using 1 when $t = 1$.

Compute the logarithm of a_q to u by Equation (2.2), partitioning H_d as necessary. In each final piece its logarithm $\ell \in [0, q^s)$ is divisible by q , because a is a q -th power. Therefore $u^{\ell/q}$ is a root of a_q on that piece. Multiply the two roots, recombine the pieces by polynomial Chinese remaindering, and finally recombine the relatively prime H_d .

For complexity, $d \leq h$, the bit length of Q^d is at most $1 + h \log_2 Q$, and $s \leq h \log_2 Q$. There are at most h nonempty pieces at every digit level; each digit tests at most q possibilities. Equation (2.4) is a linear system of dimension at most h , and Equation (2.5) uses at most $h^2 + 1$ trials. All remaining operations are polynomial arithmetic, powering, or integer Euclidean algorithms of the indicated sizes. This proves the claimed polynomial bound and also the assertion about field instructions. $\square$

We denote this procedure by **UnitRoot**. To extract a root in k itself, apply it with the linear modulus $H = X$. The proof explains why distinct-degree decomposition suffices: every subsequent partition is obtained by an equality test, and full irreducible factorization is never invoked.

2.4 Splitting an even number of roots

Rónyai proved that a supplied quadratic nonresidue suffices to split a completely split polynomial of even degree in polynomial time [25]. A quadratic nonresidue gives a full 2-primary generator by raising it to the odd part of $p - 1$. We give the pair-orientation argument in the form needed here. A primary generator forces a distinction among an even number of roots. The mechanism is an orientation of the pairs of distinct roots: each pair contributes one to precisely one of its two row counts. Equal row counts would then have the nonintegral value $(N - 1)/2$.

Proposition 2.3 (Even-degree splitting). *Let p be odd, and let $F \in \mathbf{F}_p[T]$ be monic, square-free, and totally split of even degree $N \geq 2$, with $p > N$. Given a generator u of the full 2-primary subgroup of $\mathbf{F}_p^*$, one can deterministically find a nontrivial proper monic divisor of F in a number of bit operations polynomial in N and $\log p$.*

Proof. Set $A = \mathbf{F}_p[T]/F$ and $R = A \otimes_{\mathbf{F}_p} A$. Write $p - 1 = 2^s t$ with t odd. In the coordinate description indexed by ordered pairs of roots, the element

$$\Delta = T \otimes 1 - 1 \otimes T$$

has coordinate $x_i - x_j$. Thus $e = \Delta^{p-1}$ is the idempotent equal to 1 off the diagonal and 0 on the diagonal. Work in the algebra eR , whose identity is e . The element Δ^t has 2-power order at every coordinate there, so compute its logarithm to u by the digit procedure. For a masked equality test on an element z in a subalgebra with identity mask e_0 , its zero mask is $e_0 - (e_0 z)^{p-1}$. Hence every partition is computed within R , with no knowledge of the roots.

Let e_* be the sum of the final masks whose logarithms lie in $[0, 2^{s-1})$. Replacing (i, j) by (j, i) changes Δ^t to its negative, because t is odd. The logarithm therefore changes by

2^{s-1} modulo 2^s , and exactly one orientation of each unordered pair is selected by e_* . Apply the ordinary algebra trace on the second factor:

$$c = (\text{id} \otimes \text{Tr}_{A/\mathbf{F}_p})(e_*) \in A. \quad (2.6)$$

Its i -th coordinate is the number c_i of selected pairs in row i , viewed in $\mathbf{F}_p$. As integers,

$$0 \leq c_i \leq N-1, \quad \sum_{i=1}^N c_i = \frac{N(N-1)}{2}.$$

If these integers were all equal, their common value would be $(N-1)/2$, which is impossible for even N . Since $p > N$, their images in the field are not all equal either. Testing $\gcd(F, c-a)$ for $a = 0, \dots, N-1$ therefore returns a nontrivial proper factor.

The algebra R has dimension N^2 , and the number of nonempty masks at a digit level is at most N^2 . There are $s \leq \log_2 p$ levels, each with two possible digits. Binary powering, the trace computation, and the final gcds all have the asserted polynomial cost. $\square$

3 Auxiliary primes and primary generators

The splitting procedure needs explicit generators of small-prime parts of multiplicative groups. This section constructs those generators from the auxiliary primes supplied in Theorem 1.2. The construction uses linear algebra in cyclotomic algebras and calls the factorization procedure only in strictly smaller degrees. Existence of small auxiliary primes is a separate number-theoretic question; no size bound for them is assumed in this section.

We use the notation $v_q(a)$ and the full q -primary subgroup introduced in Section 2.

Throughout the construction, $p > n \geq 2$. For each prime $q \leq n$, recall that the supplied auxiliary prime ℓ_q satisfies

$$\ell_q \notin \{2, 3, p, q\}, \quad \ell_q \equiv 1 \pmod{12q}, \quad p^{(\ell_q-1)/q} \not\equiv 1 \pmod{\ell_q}. \quad (3.1)$$

Recall the bound $E = \max(2, \max_{q \leq n} \ell_q)$ from Theorem 1.2. Supplied primes can be checked by trial division and modular exponentiation in time polynomial in E , n , and $\log p$. The same tests can search upward for the first suitable prime; if one exists below E , that search also takes time polynomial in these parameters. Factoring $\ell_q - 1$ is unnecessary.

The table has the following contents:

- for $q = 2$, an element $\omega_2 \in \mathbb{F}_p^*$ of order $2^{v_2(p-1)}$;
- for each odd prime $q \leq n$, an explicitly represented field $K_q/\mathbb{F}_p$, an element $\zeta_q \in K_q$ of order q with $K_q = \mathbb{F}_p(\zeta_q)$, and an element $\omega_q \in K_q^*$ of order $q^{v_q(|K_q|-1)}$.

An explicit field here consists of an $\mathbb{F}_p$ -basis, its multiplication constants, and the coordinates of its identity. A quotient by a specified irreducible polynomial is an equivalent representation. Both descriptions support arithmetic and linear algebra in polynomial time. We use the standard structure theory of finite fields; see [17].

3.1 A degree- q field without factoring a cyclotomic polynomial

Fix q and $\ell = \ell_q$. Put

$$R_\ell = \mathbb{F}_p[T]/(1 + T + \dots + T^{\ell-1}), \quad \Gamma_\ell = (\mathbb{Z}/\ell\mathbb{Z})^*, \quad H_q = \{a^q : a \in \Gamma_\ell\}.$$

For $a \in \Gamma_\ell$, let τ_a be the automorphism of R_ℓ defined by substitution $T \mapsto T^a$. Enumerate H_q by taking q th powers modulo ℓ , and compute

$$U_q = R_\ell^{H_q} = \bigcap_{a \in H_q} \ker(\tau_a - \text{id}) \quad (3.2)$$

by $\mathbb{F}_p$ -linear algebra. Products of basis elements of this space, reduced in R_ℓ and expressed in its computed basis, give the multiplication constants of U_q .

Lemma 3.1. *The algebra U_q is a field of degree q over $\mathbb{F}_p$.*

Proof. Since $p \neq \ell$, the polynomial defining R_ℓ has $\ell - 1$ distinct roots over an algebraic closure $\overline{\mathbb{F}}_p$. They are precisely the primitive ℓ th roots of unity. Evaluation at these roots identifies the scalar extension of R_ℓ with the algebra of functions on that set. The group Γ_ℓ acts regularly on the set, and its subgroup H_q has index q . Taking kernels commutes with extension of scalars. Consequently

$$U_q \otimes_{\mathbb{F}_p} \overline{\mathbb{F}}_p \simeq \overline{\mathbb{F}}_p^q,$$

with the factors indexed by the H_q -orbits.

Frobenius permutes the primitive roots by raising them to the p th power, hence acts on the orbit set by multiplication by the class of p in Γ_ℓ/H_q . A cyclic group of order $\ell - 1$ has $H_q = \{a : a^{(\ell-1)/q} = 1\}$, so the last condition in (3.1) says that this class is nonidentity. The quotient has prime order q ; its nonidentity elements act transitively on it. A finite reduced algebra over a finite field is a product of finite fields, and its field factors correspond exactly to Frobenius orbits on its geometric points. There is one such orbit, of length q , proving the claim. $\square$

This construction is useful precisely because the computation of U_q never asks for the individual irreducible factors of the degree $\ell - 1$ cyclotomic polynomial. Its field property follows from the verified power-residue test on the integer prime ℓ .

3.2 Extracting the primary generators

For $q = 2$, solve the $\mathbb{F}_p$ -linear equation

$$v^p = -v \quad (v \in U_2)$$

and take a nonzero solution. The trace map $v \mapsto v + v^p$ from the quadratic field U_2 to $\mathbb{F}_p$ is nonzero, because it sends 1 to 2. Its kernel has dimension one, so this instruction succeeds. Then $c = v^2$ lies in $\mathbb{F}_p^*$ and

$$c^{(p-1)/2} = v^{p-1} = -1.$$

Thus c is a nonsquare. With $s = v_2(p - 1)$, set

$$\omega_2 = c^{(p-1)/2^s}. \quad (3.3)$$

The order of a nonsquare in a cyclic group of even order has the full 2-part of that order. The odd exponent in (3.3) therefore leaves ω_2 of exact order 2^s .

For odd q , first factor

$$\Phi_q(Z) = 1 + Z + \cdots + Z^{q-1}$$

over $\mathbb{F}_p$. The smaller-degree nature of this call will be justified below. Choose the first monic irreducible factor in a fixed coefficient order, form its quotient field K_q , and let ζ_q

be the image of Z . Every root of Φ_q has order q , because $p \neq q$. Writing $d_q = [K_q : \mathbb{F}_p]$, finite-field theory gives

$$d_q = \text{ord}_q(p) \mid q - 1,$$

where $\text{ord}_q(p)$ is the multiplicative order of p modulo q . In particular $\gcd(d_q, q) = 1$. The tensor product

$$V_q = U_q \otimes_{\mathbb{F}_p} K_q$$

is therefore a field of degree q over K_q : finite fields of coprime degrees intersect in $\mathbb{F}_p$, so they are linearly disjoint. Its multiplication table is obtained directly from those of its two factors.

Set $Q_q = |K_q|$ and $s_q = v_q(Q_q - 1)$. Solve the K_q -linear system

$$u^{Q_q} = \zeta_q u \quad (u \in V_q) \quad (3.4)$$

and choose a nonzero solution. To prove existence, the homomorphism $x \mapsto x^{Q_q-1}$ on V_q^* has image equal to the norm-one subgroup for V_q/K_q . Indeed both subgroups have order $(Q_q^q - 1)/(Q_q - 1)$. The norm of ζ_q equals $\zeta_q^{1+Q_q+\dots+Q_q^{q-1}} = \zeta_q^q = 1$. Thus ζ_q lies in this image, proving that (3.4) has a nonzero solution.

For odd q and $Q_q \equiv 1 \pmod{q}$, the lifting-the-exponent identity is

$$v_q(Q_q^a - 1) = s_q + v_q(a) \quad (a \geq 1). \quad (3.5)$$

This is Equation (2.3). In particular $v_q(Q_q^q - 1) = s_q + 1$. Equation (3.4) says that u^{Q_q-1} has order exactly q , so the q -part of the order of u is exactly q^{s_q+1} . Consequently

$$w = u^{(Q_q^q-1)/q^{s_q+1}}, \quad \omega_q = w^q$$

have orders q^{s_q+1} and q^{s_q} , respectively. The unique subgroup of V_q^* of order q^{s_q} lies in K_q^* , since $q^{s_q} \mid Q_q - 1$. Therefore $\omega_q \in K_q$, as required. Its coordinates in K_q are obtained by solving the linear equations for the known inclusion $K_q \hookrightarrow V_q$.

All integer valuations used here are found by repeated division by the specified prime q . All powers use binary exponentiation. Neither construction requires an integer factorization of $p - 1$ or $Q_q - 1$.

3.3 The increasing-degree construction

The only call above to an as yet unspecified factoring procedure is the factorization of Φ_q , whose degree is $q - 1$. The interface with the rest of the paper is the following.

Proposition 3.2 (Table construction). *Suppose the auxiliary primes (3.1) are supplied, and suppose a deterministic procedure has the following uniform contract: for every integer $1 \leq b < n$, it factors every monic polynomial of degree at most b over $\mathbb{F}_p$ using table entries only for primes at most b . Then the complete table through n can be built in increasing prime order. Its construction calls that procedure at most once for each odd prime $q \leq n$, on a polynomial of degree $q - 1$; each such call uses only already constructed table entries.*

Apart from these factorization calls, the construction has bit cost polynomial in E , n , and $\log p$, with an absolute exponent. Each stored odd-prime field has degree at most $q - 1$, and the temporary field V_q has degree at most $q(q - 1)$ over $\mathbb{F}_p$.

Proof. Begin with the quadratic construction, which needs no factorization call. For an odd prime q , every prime at most $q - 1$ is smaller than q , so the indicated factorization call has all its required entries. Its output constructs K_q , and the preceding linear-algebra construction supplies ω_q . This proves the inductive assertion and the degree bounds.

For completeness, the largest auxiliary algebra R_{ℓ_q} has dimension $\ell_q - 1 \leq E$. Substitution operators are computed by modular polynomial powering. There are at most $\ell_q - 1$ elements in H_q , and their fixed-space equations have at most $\ell_q - 1$ unknowns. Gaussian elimination gives its invariant subspace and its multiplication constants in polynomial time. All subsequent field dimensions are at most n^2 . Including the modular residue tests, the exponent integers used have $O(\log(E + 1) + n^2 \log p)$ bits, and each extraction of a specified prime valuation has polynomial cost. Schoolbook arithmetic consequently gives, for example, an upper bound

$$O(n(E + 1)^{10}(n + \lceil \log_2 p \rceil + 1)^{40})$$

for the work outside the factorization calls. The same generous bound includes trial division and upward searches through E . $\square$

Section 9 proves that FACTOR has exactly this interface. Its correctness and the table construction are established by a common induction: the entry for q invokes factorization only below q , whereas factorization in degree b invokes entries only at most b . Thus the table uses no factorization oracle. The only external data in this construction are the verified auxiliary primes.

4 Divisions on a cyclic cover

The odd-degree splitting procedure will repeatedly divide divisor classes by $1 - \sigma$, where σ is a cyclic automorphism of a curve. Its first requirement is an existence statement: all the needed divisions must be rational over an extension of polynomial degree. Its second requirement is a test that eventually distinguishes two ramification points. We establish the geometric inputs here. The essential device is to retain a divisor supported at infinity together with a chosen division of its class in the Jacobian. Equality of classes may lose information about that divisor; retaining the divisor makes a final divisibility obstruction visible.

4.1 The curve and its infinity lattice

Let K be a finite field of characteristic p , let $q \neq p$ be an odd prime, and suppose that K contains a primitive q -th root of unity ζ . Let $F \in K[X]$ be monic and square-free, of degree $N \geq q$ divisible by q , and suppose that all its roots $x_1, \dots, x_N$ belong to K . Consider the smooth projective curve C/K with function field

$$K(C) = K(X, Y), \quad Y^q = F(X).$$

The roots x_i serve only as mathematical indices here; their values need not be known to an algorithm.

A simple root of F has valuation one, so F is not a q -th power even over $\overline{K}(X)$. Thus C is geometrically integral and the map $X : C \rightarrow \mathbb{P}^1$ has degree q . The affine equation is smooth: a singularity would have $Y = 0$ and $F(X) = F'(X) = 0$. At infinity, put

$$z = X^{-1}, \quad U = Yz^{N/q}, \quad U^q = z^N F(z^{-1}).$$

At $z = 0$ the right side is 1, and its q roots are simple. There are therefore exactly q rational points at infinity,

$$I = \{\infty_j : 0 \leq j < q\}, \quad U(\infty_j) = \zeta^j.$$

The ramification points are $P_i = (x_i, 0)$. Each is totally and tamely ramified, and there is no other ramification. The tame Riemann–Hurwitz formula [29, Tags 0C1B and 0C1F] consequently gives

$$g(C) = \frac{(q-1)(N-2)}{2}. \quad (4.1)$$

Let σ send (X, Y) to $(X, \zeta Y)$ on points. We let it act on functions by

$$(\sigma h)(X, Y) = h(X, \zeta^{-1} Y), \quad (4.2)$$

so that $\operatorname{div}(\sigma h) = \sigma \operatorname{div}(h)$. In particular, $\sigma P_i = P_i$ and $\sigma \infty_j = \infty_{j+1}$, with subscripts taken modulo q .

Write $J = \operatorname{Pic}^0(C_{\overline{K}})$ for the group of geometric degree-zero divisor classes, and put

$$\Lambda = \left\{ \sum_{j=0}^{q-1} a_j \infty_j : a_j \in \mathbb{Z}, \sum_j a_j = 0 \right\}.$$

For a degree-zero divisor D , its class is denoted by $[D]$. The orbit sum $1 + \sigma + \cdots + \sigma^{q-1}$ acts as zero on both J and Λ . For J , this follows because the orbit sum of a divisor is the pullback of its pushforward to $\mathbb{P}^1$, and a degree-zero divisor on $\mathbb{P}^1$ is principal. For Λ , it follows by summing its coefficients.

Let $R = \mathbb{Z}[\xi]$, where ξ is an abstract primitive q -th root of unity, and let ξ act as σ . The preceding orbit-sum identity makes J and Λ into R -modules. Set

$$\lambda = 1 - \xi, \quad e = (\sigma - 1)\infty_0. \quad (4.3)$$

The differences $\sigma^j e$ generate Λ . The resulting map $R \rightarrow \Lambda$, $a \mapsto ae$, is an isomorphism: it is surjective between free abelian groups of the same rank $q-1$. In particular, multiplication by λ is injective on Λ . On divisors and divisor classes, multiplication by λ means $1 - \sigma$.

We shall use

$$q = u\lambda^{q-1} \quad \text{for some } u \in R^*, \quad R/(\lambda) = \mathbb{F}_q. \quad (4.4)$$

Indeed, $q = \prod_{a=1}^{q-1} (1 - \xi^a)$, and each quotient $(1 - \xi^a)/(1 - \xi)$ is an algebraic integer of norm one, hence a unit. The quotient-ring assertion follows by setting $\xi = 1$ in the cyclotomic polynomial $1 + T + \cdots + T^{q-1}$. Multiplication by λ is surjective on J . To see this, identify J with the geometric points of the Jacobian of C [19, Theorem 1.1]. Multiplication by q , which is prime to the characteristic, is a surjective isogeny [18, Theorem 8.2]. Equation (4.4) then implies the surjectivity of multiplication by λ .

4.2 Recording a class together with an infinity divisor

Ramification divisors and the deck-transformation operator $1 - \sigma$ also underlie explicit descent on Jacobians of cyclic covers; see Poonen and Schaefer [24, Section 6, Proposition 6.2]. Here we record the infinity divisor as an additional coordinate, so that successive divisions retain its integral divisibility information.

For $t \geq 1$, define the R -module

$$T_t = \frac{\{(d, W) \in J \times \Lambda : \lambda^t d = [W]\}}{\{([W'], \lambda^t W') : W' \in \Lambda\}}. \quad (4.5)$$

A pair in the numerator will also denote its image in the quotient when there is no ambiguity. The quotient identifies changes obtained by adding an infinity divisor to a representative of d . Keeping W itself, instead of only its class, will allow us to test divisibility in the free module Λ .

Lemma 4.1. *The module T_t is killed by λ^t . The map*

$$T_t \longrightarrow T_{t+1}, \quad (d, W) \longmapsto (d, \lambda W)$$

is injective, and its image is $T_{t+1}[\lambda^t]$. Consequently, inside any T_m the successive modules are precisely $T_j = T_m[\lambda^j]$ for $1 \leq j \leq m$.

Proof. For a qualifying pair, multiplication by λ^t gives $([W], \lambda^t W)$, which is one of the quotient relations. The displayed map respects both the qualifying equation and the relations. If its value is zero, then

$$(d, \lambda W) = ([W'], \lambda^{t+1} W')$$

for some $W' \in \Lambda$. Injectivity of λ on the lattice gives $W = \lambda^t W'$, so the original pair was already zero in T_t . Conversely, suppose that the class of (d, W) in T_{t+1} is killed by λ^t . There is then a $W' \in \Lambda$ such that

$$\lambda^t d = [W'], \quad \lambda^t W = \lambda^{t+1} W'.$$

The second equation gives $W = \lambda W'$, and (d, W) is the image of $(d, W') \in T_t$. Iteration proves the last assertion. $\square$

The first module has an explicit description in terms of the ramification points. This description will serve twice: it controls Frobenius on all higher modules, and it provides the labels used to distinguish roots.

Lemma 4.2 (Ramification generators and labels). *In the setup of Section 4.1, the elements*

$$\epsilon_i = ([P_i - \infty_0], e) \quad (1 \leq i \leq N)$$

generate T_1 over $\mathbb{F}_q$, and

$$\sum_{i=1}^N \epsilon_i = 0. \tag{4.6}$$

More explicitly, let D be a geometric degree-zero divisor, let $W \in \Lambda$, and suppose that

$$\operatorname{div}(h) = (1 - \sigma)D - W.$$

Then $\operatorname{Norm}_{\overline{K}(C)/\overline{K}(X)}(h)$ is a nonzero constant. Choose $\gamma \in \overline{K}^$ with*

$$\gamma^q = \operatorname{Norm}_{\overline{K}(C)/\overline{K}(X)}(h).$$

Then $h(P_i)$ is defined and nonzero, and the residues

$$l_i = v_{P_i}(D) - \log_\zeta(h(P_i)/\gamma) \quad \text{in } \mathbb{F}_q \tag{4.7}$$

satisfy

$$([D], W) = \sum_{i=1}^N l_i \epsilon_i \quad \text{in } T_1. \tag{4.8}$$

Here $v_{P_i}(D)$ is the coefficient of P_i in D , and $\log_\zeta$ is the unique exponent modulo q of a q -th root of unity.

Proof. Since $\lambda[P_i - \infty_0] = [e]$, each ϵ_i qualifies. The divisor of Y is

$$\operatorname{div}(Y) = \sum_i P_i - \frac{N}{q} \sum_j \infty_j.$$

Thus, with

$$W' = \frac{N}{q} \sum_j \infty_j - N\infty_0,$$

we have $\sum_i [P_i - \infty_0] = [W']$ and $\lambda W' = Ne$. This is exactly the relation proving (4.6).

Now take any pair $([D], W)$ in the numerator defining T_1 , and choose h with the stated divisor. The orbit sum of $(1 - \sigma)D - W$ is zero, so the norm of h has zero divisor on $\mathbb{P}^1$ and is a nonzero constant. Hilbert's Theorem 90 [9, Example 2.3.4] for the cyclic extension $\overline{K}(C)/\overline{K}(X)$ gives

$$h/\gamma = b/(\sigma b) \quad (4.9)$$

for some $b \in \overline{K}(C)^*$. Choose an integral divisor W_0 supported on I such that $(1 - \sigma)W_0 = W$. Such a divisor exists because the image of $1 - \sigma$ on the full permutation lattice $\mathbb{Z}^I$ is its augmentation lattice Λ . It need not have degree zero. Taking divisors in (4.9) shows that

$$D - \operatorname{div}(b) - W_0$$

is invariant under σ .

An invariant divisor has equal coefficients along each unramified orbit. At a ramification point its coefficient may be reduced modulo q , because pullback multiplies that coefficient by q . Hence

$$D - \operatorname{div}(b) - W_0 = \sum_i a_i P_i + X^* D_0, \quad 0 \leq a_i < q, \quad (4.10)$$

for a divisor D_0 on $\mathbb{P}_K^1$. Put

$$W' = W_0 + \left(\sum_i a_i \right) \infty_0 + (\deg D_0) \sum_j \infty_j.$$

Taking degrees in (4.10) gives $\deg W' = 0$. A divisor of degree $\deg D_0$ on $\mathbb{P}^1$ is linearly equivalent to $(\deg D_0)\infty$, so

$$[D] = [W'] + \sum_i a_i [P_i - \infty_0], \quad \lambda W' = W - \sum_i a_i e.$$

The pair $([W'], \lambda W')$ vanishes in T_1 , proving generation and the expansion with coefficients a_i modulo q .

It remains to identify these coefficients from D and h . At P_i , both $(1 - \sigma)D$ and W have coefficient zero. Thus h is a unit there. Since σ fixes P_i , evaluation of the norm gives

$$h(P_i)^q = \operatorname{Norm}(h).$$

In particular $h(P_i)/\gamma$ is a power of ζ . The function Y is a local parameter at P_i . If $v_{P_i}(b) = a$, the action (4.2) gives

$$(b/(\sigma b))(P_i) = \zeta^a.$$

On the other hand, taking the coefficient of P_i in (4.10) gives $a_i \equiv v_{P_i}(D) - a \pmod{q}$. Equation (4.9) now yields (4.7) and (4.8). $\square$

In particular, if all the labels in (4.7) agree, then $([D], W) = 0$ in T_1 . This implication does not require the labels to be unique. Replacing γ by another q -th root shifts all of them by the same constant, which has no effect because of (4.6).

4.3 Frobenius and rational divisions

Let Π denote the $|K|$ -power Frobenius acting on geometric points, divisors, and divisor classes. It commutes with σ and fixes Λ pointwise. Lemma 4.2 shows that it also fixes T_1 pointwise, since all P_i and all infinity points are K -rational. We next propagate this information through the filtration.

Lemma 4.3. *For the curve and modules above, let $r \geq 0$, set $m = 1 + (q - 1)r$, and let k/K be the extension of degree q^r . Then Π^{q^r} fixes T_m pointwise. If $1 \leq t \leq m$ and $d \in J$, $W \in \Lambda$ satisfy $\lambda^t d = [W]$, then d itself is fixed by the Frobenius of k .*

Proof. On T_m , let $F_j = \ker(\lambda^j)$ for $0 \leq j \leq m$, and put $F_j = 0$ for negative j . By Lemma 4.1, $F_1 = T_1$. For $1 \leq j \leq m$ and $v \in F_j$,

$$\lambda^{j-1}(\Pi - 1)v = (\Pi - 1)\lambda^{j-1}v = 0.$$

Thus $\Pi - 1$ lowers the filtration by one. Multiplication by q lowers it by $q - 1$, by (4.4).

Suppose that an endomorphism $A = \Pi^{q^a} - 1$ lowers the filtration by at least $f \geq 1$. The binomial expansion gives

$$\Pi^{q^{a+1}} - 1 = qA + \binom{q}{2}A^2 + \cdots + \binom{q}{q-1}A^{q-1} + A^q.$$

Every intermediate coefficient is divisible by q . Its term therefore lowers by at least $f + q - 1$; the last term lowers by $qf \geq f + q - 1$. Induction shows that $\Pi^{q^r} - 1$ lowers by $1 + r(q - 1) = m$ and hence vanishes on T_m .

For the final assertion, T_t embeds in T_m , so the class of (d, W) in T_t is fixed. The lattice component is unchanged. Consequently, for some $W' \in \Lambda$,

$$(\Pi^{q^r} d - d, 0) = ([W'], \lambda^t W').$$

The second coordinate and injectivity of λ on Λ force $W' = 0$. The first coordinate then says $\Pi^{q^r} d = d$. $\square$

For computation we need divisors and functions over k , not just fixed geometric classes. The following finite-field descent fact supplies them. We include its proof to make the distinction explicit.

Lemma 4.4 (Finite-field descent). *Let C be a smooth projective geometrically integral curve over a finite field k . A k -rational divisor that is geometrically principal is the divisor of a function in $k(C)^*$. Every geometric divisor class fixed by the Frobenius of k has a k -rational divisor representative.*

Proof. For the first assertion, choose a function h with the given divisor over a finite constant extension k'/k . If τ generates $\text{Gal}(k'/k)$, then $\tau h/h$ is a constant in $(k')^*$, and its norm to k is one. Hilbert's Theorem 90 for k'/k lets us scale h by a constant so that it is fixed by τ .

For the second assertion, choose a representative D over some finite extension k'/k . Since its class is fixed, $D - \tau D$ is principal. The first assertion, applied over k' , supplies $h \in k'(C)^*$ with

$$D - \tau D = \text{div}(h).$$

The norm of h in the constant-field extension has zero divisor, hence is a constant in k^* . The norm $(k')^* \rightarrow k^*$ is surjective, so a constant multiple of h has norm one. Hilbert's Theorem 90 for $k'(C)/k(C)$ now gives $h = b/(\tau b)$ after that scaling. It follows that $D - \text{div}(b)$ is invariant under τ . An invariant divisor descends by grouping its points into Galois orbits, and its class is the class of D . $\square$

Corollary 4.5. *In the setup of Section 4.1, let $r \geq 0$, $m = 1 + (q - 1)r$, and $[k : K] = q^r$. For every i there exist classes $d_{i,1}, \dots, d_{i,m}$, all represented by degree-zero divisors over k , such that*

$$\begin{aligned} d_{i,1} &= [P_i - \infty_0], \\ \lambda d_{i,t+1} &= d_{i,t} \quad (1 \leq t < m), \\ \lambda^t d_{i,t} &= [e] \quad (1 \leq t \leq m). \end{aligned} \tag{4.11}$$

Moreover, every successive choice of a geometric division in such a chain has a representative over k .

Proof. Surjectivity of λ on J permits each successive choice over $\overline{K}$. The identity $\lambda d_{i,1} = [e]$ and induction give the last equation in (4.11). Thus $(d_{i,t}, e)$ qualifies for T_t . Lemma 4.3 fixes $d_{i,t}$ over k , and Lemma 4.4 supplies its divisor representative over k . The argument applies to every chosen division, not merely to a special compatible family. $\square$

4.4 Why the labels must separate ramification points

Set $r = v_q(N)$ in Corollary 4.5 and sum the last members of its lift families. The resulting class $d_m = \sum_i d_{i,m}$ satisfies $\lambda^m d_m = [Ne]$. This single relation is enough to force a distinction between ramification points; the particular choices of lifts no longer matter. We now prove that assertion before turning to the algorithms that construct the divisors and evaluate their labels.

Proposition 4.6 (Forced separation). *In the setup of Section 4.1, put $r = v_q(N)$ and $m = 1 + (q - 1)r$, and let k/K have degree q^r . Let $d_m \in J$ have a degree-zero divisor representative over k and satisfy $\lambda^m d_m = [Ne]$. For $1 \leq t \leq m$, set*

$$d_t = \lambda^{m-t} d_m, \quad W_t = \lambda^{-(t-1)} Ne \in \Lambda, \tag{4.12}$$

and choose any degree-zero divisor D_t over k representing d_t . The lattice divisions defining W_t exist and are unique.

Starting at $t = 1$, make the following test, advancing to $t + 1$ only when all labels at t agree. Choose $h_t \in k(C)^*$ and $\gamma_t \in k^*$ such that

$$\operatorname{div}(h_t) = (1 - \sigma)D_t - W_t, \quad \gamma_t^q = \operatorname{Norm}(h_t), \tag{4.13}$$

and form the labels

$$l_{i,t} = v_{P_i}(D_t) - \log_{\zeta}(h_t(P_i)/\gamma_t) \quad \text{in } \mathbb{F}_q. \tag{4.14}$$

At every reached test these choices exist over k and the labels are defined. For some reached $t \leq m$, the labels are not all equal. This conclusion holds for every choice of the representatives D_t and of the functions and roots in (4.13).

Proof. Write $N = q^r a$ with $q \nmid a$. By (4.4), q^r is an associate of λ^{m-1} , whereas the image of a in $R/(\lambda) = \mathbb{F}_q$ is nonzero. Thus

$$Ne \in \lambda^{m-1} \Lambda \setminus \lambda^m \Lambda. \tag{4.15}$$

This proves that all the W_t in (4.12) exist uniquely, and that $W_m \notin \lambda \Lambda$.

The relation $\lambda d_t = [W_t]$ is not asserted at every level in advance. It is the invariant that permits a test to be reached. For $t = 1$, it follows from $\lambda^m d_m = [Ne]$. Suppose that test t is reached with this invariant. Then $(1 - \sigma)D_t - W_t$ is a principal divisor over k . Lemma 4.4

supplies $h_t \in k(C)^*$ with this divisor. Lemma 4.2 shows that h_t is a unit at every P_i and that

$$\text{Norm}(h_t) = h_t(P_i)^q \in k^*.$$

Hence a root γ_t exists in k , and every ratio $h_t(P_i)/\gamma_t$ is a q -th root of unity. The labels are defined and satisfy

$$(d_t, W_t) = \sum_i l_{i,t} \epsilon_i \quad \text{in } T_1.$$

If these labels are not all equal, the conclusion holds. If they agree, Equation (4.6) makes this pair zero. The quotient defining T_1 therefore gives an actual infinity divisor $W' \in \Lambda$ with

$$d_t = [W'], \quad W_t = \lambda W'. \quad (4.16)$$

For $t < m$, injectivity of λ on Λ identifies W' with W_{t+1} . Since $\lambda d_{t+1} = d_t$, we obtain $\lambda d_{t+1} = [W_{t+1}]$, exactly the invariant for the next test. At $t = m$, however, (4.16) contradicts $W_m \notin \lambda\Lambda$. Not all tests can have equal labels. The argument used no restrictions on the representatives or the choices in (4.13), proving the final assertion. $\square$

Figure 2 separates the two traversals in this proof. The class d_m determines the class chain by applying λ ; the divisor Ne determines the infinity chain by dividing by λ . The label tests compare their progress and would force one impossible extra lattice division if every test were constant.

$$\begin{array}{ccccccc}
 J & & d_1 & \xleftarrow{\lambda} & d_2 & \xleftarrow{\lambda} & \cdots & \xleftarrow{\lambda} & d_m \\
 \\
 \Lambda & & W_1 = Ne & \xrightarrow{\text{divide by } \lambda} & W_2 & \xrightarrow{\text{divide by } \lambda} & \cdots & \xrightarrow{\text{divide by } \lambda} & W_m \\
 & & & & & & & & W_m \notin \lambda\Lambda
 \end{array}$$

Figure 2: The two chains in Proposition 4.6. Tests run from left to right. Reaching test t requires $\lambda d_t = [W_t]$; equal labels at $t < m$ give $d_t = [W_{t+1}]$ and allow the next test. Equal labels at the final test would require the forbidden additional division of W_m in Λ . No equality between the rows is assumed before its test is reached.

The geometric mechanism is now complete: Corollary 4.5 supplies the class d_m , and Proposition 4.6 forces distinct labels using that class. The next sections implement divisor arithmetic, class division, and evaluation without recovering the roots of F or factoring the finite supports of divisors.

5 Divisor arithmetic without factoring supports

The lifting argument uses rational divisors, although the finite points in their supports are not given individually. We now give a representation in which addition, reduction, and evaluation require only polynomial arithmetic and linear algebra. Two further operations will be needed: summing divisors computed simultaneously at the unknown roots of F , and evaluating a principal function whose divisor has very large coefficients at infinity. Linear-algebra algorithms for Riemann–Roch spaces and divisor arithmetic are established tools; see Hess [13] and Khuri-Makdisi [16]. We give the representation and size bounds explicitly because both the genus and the degree of the cover vary in this application.

Throughout this section, k is a finite field, q is an odd prime different from its characteristic, k contains a specified primitive q th root ζ , and $F \in k[X]$ is monic and square-free of degree $N \geq q$, with $q \mid N$. We use the curve, automorphism, and rational points at infinity introduced above. Thus

$$C : Y^q = F(X), \quad \mathcal{L} = k(C), \quad O = k[X, Y]/(Y^q - F(X)), \quad g = \frac{(q-1)(N-2)}{2}.$$

The automorphism σ sends points by $Y \mapsto \zeta Y$ and acts on functions by $Y \mapsto \zeta^{-1}Y$. All divisors in the computation are k -rational divisors, written as sums of closed points. For a divisor D , write

$$\|D\| = \sum_P |v_P(D)| [k(P) : k]$$

for its absolute degree. Bounds polynomial in $\|D\|$ do not mean bounds polynomial in the bit length of arbitrary divisor coefficients. Large coefficients supported at infinity are treated separately at the end of this section.

Proposition 5.1 (Effective divisor arithmetic). *On the curve C above, the following operations are deterministic and use a number of field operations polynomial in q, N , the degrees occurring in the input representation, the absolute degrees of the divisors materialized during the operation, and the parameter ℓ in the third operation:*

1. divisor addition, negation, coefficientwise minimum and maximum, and the action of σ ;
2. construction of the divisor of an explicitly represented nonzero function, and evaluation of its valuation and leading coefficient at a specified ramification point or point at infinity;
3. computation of the Riemann–Roch space $L(D + \ell\infty_0)$ for $\ell \geq 0$;
4. replacement of a degree-zero divisor D by a linearly equivalent divisor D' satisfying $\|D'\| \leq 2g$, with a function a such that $D' = D + \text{div}(a)$.

These algorithms do not require factorization of the polynomials defining the finite supports. Their decisions use only field zero tests, so they admit simultaneous execution as in Lemma 2.1.

We prove the proposition by constructing the algorithms. The final two subsections extend them to simultaneous sums and to binary infinity coefficients.

5.1 Fractional ideals and finite linear algebra

Smoothness of the affine curve implies that O is a Dedekind domain. It is a free $k[X]$ -module with basis $1, Y, \dots, Y^{q-1}$. The finite part of a divisor D is represented by the fractional ideal

$$I(D) = \{a \in \mathcal{L} : v_P(a) \geq v_P(D) \text{ at every finite point } P\}. \quad (5.1)$$

The zero function is included in this definition. The rest of the representation is the integer vector $(v_{\infty_0}(D), \dots, v_{\infty_{q-1}}(D))$.

To store a fractional ideal I , choose a monic polynomial $H \in k[X]$ such that

$$HO \subseteq I \subseteq H^{-1}O. \quad (5.2)$$

Store the image of I as a k -subspace of $H^{-1}O/HO$. Multiplication by H identifies this quotient with O/H^2O , of dimension $2q \deg H$. Thus a subspace basis consists of polynomial vectors modulo H^2 in the standard Y -basis. Lifting this basis and adjoining H gives a finite

list of O -generators of I . Conversely, from such generators and a bound (5.2), take their images and repeatedly enlarge their k -span under multiplication by X and Y . The process stabilizes in at most $2q \deg H$ strict enlargements and gives the required ideal image. The convention $H = 1$ represents $I = O$ with a zero-dimensional quotient.

The identities

$$\begin{aligned} I(D + E) &= I(D)I(E), & I(-D) &= I(D)^{-1}, \\ I(\min(D, E)) &= I(D) + I(E), & I(\max(D, E)) &= I(D) \cap I(E) \end{aligned}$$

reduce the first group of operations to finite linear algebra. For a product, take pairwise products of generators and use the bound $H_D H_E$. For a sum or intersection, embed both subspaces using that same common bound, and take their sum or intersection. For inversion, retain the bound H : an element $a \in H^{-1}O$ belongs to I^{-1} precisely when $au \in O$ for every generator u of I . These are linear conditions on $H^{-1}O/HO$. They are independent of the chosen representative of a , since $HI \subseteq O$. The action of σ is substitution of $\zeta^{-1}Y$ for Y in the generators. The corresponding operations on the infinity vectors are immediate.

A known ramification point $P = (x, 0)$ has ideal $(X - x, Y)$, with bound $H = X - x$. For a nonzero function a , arithmetic in $\mathcal{L}$ uses rational coefficients in $k(X)$ in the standard Y -basis. Inversion is linear algebra over $k(X)$. Clearing denominators in both a and a^{-1} gives a polynomial H for which aO satisfies (5.2); hence the finite part of $\text{div}(a)$ is computable without finding any of its points.

Intermediate bound polynomials need not be minimal. The following compression both controls their degrees and supplies the norm data used in class division.

Lemma 5.2 (Compression and pushforward). *From the representation of $I = I(D)$ one can compute monic polynomials $h^+, h^- \in k[X]$ satisfying*

$$\deg h^+ = \sum_{P \text{ finite}} \max(v_P(D), 0)[k(P) : k], \quad \deg h^- = \sum_{P \text{ finite}} \max(-v_P(D), 0)[k(P) : k].$$

The polynomial $H = h^+ h^-$ satisfies (5.2). If $\deg D = 0$ and $j = h^+/h^-$, then

$$\text{div}_C(j) = \sum_{a=0}^{q-1} \sigma^a D. \tag{5.3}$$

Proof. Form the integral ideals $I^+ = I \cap O$ and $I^- = I^{-1} \cap O$. Let h^+ and h^- be the characteristic polynomials of multiplication by X on O/I^+ and O/I^- , respectively. These quotients are computed from the stored subspaces; the original bound H_0 has $H_0 O \subseteq I^+, I^-$, so it suffices to work in $O/H_0 O$. A quotient at a closed point P of multiplicity d has a filtration with d successive residue fields $k(P)$. Its k -dimension is therefore $d[k(P) : k]$, proving the degree formulas.

Cayley–Hamilton gives $h^+ O \subseteq I^+ \subseteq I$ and $h^- O \subseteq I^- \subseteq I^{-1}$. The second inclusion implies $I \subseteq (h^-)^{-1} O$. Both sides of (5.2) now follow for $H = h^+ h^-$. Recompute the ideal subspace with this bound after each operation. In particular, its dimension is at most twice q times the absolute degree of the finite divisor.

For the last assertion, let $Q = X(P)$ be a finite point downstairs and f_Q its monic defining polynomial. On $k(P)$, multiplication by X has characteristic polynomial $f_Q^{[k(P):k(Q)]}$. Applying this fact to the preceding filtrations shows that the finite divisor of j on $\mathbb{P}^1$ is $X_* D$ at the finite points. Since $\deg D = 0$, the infinity coefficient agrees as well. Finally, for this cyclic cover, $X^* X_* D = \sum_{a=0}^{q-1} \sigma^a D$, including the ramified points and points with nontrivial residue extension. This proves (5.3). $\square$

5.2 Local expansions and Riemann–Roch reduction

The ideal representation controls all finite points collectively. At the few specified points where a value or inequality is required, local expansions give the additional information.

At infinity use $z = X^{-1}$ and $U = Yz^{N/q}$. The equation

$$U^q = z^N F(z^{-1}), \quad U(0) = \zeta^j$$

determines the expansion at ∞_j by simple-root Hensel lifting. At $P = (x, 0)$ use Y as uniformizer and solve $F(X) = Y^q$ with $X(0) = x$; the derivative $F'(x) \neq 0$ again permits coefficient-by-coefficient lifting. The valuation and leading coefficient of a nonzero function are then found by substitution and division of the leading terms.

Here is an explicit precision bound. Write a function as $A(X, Y)/B(X)$, where $\deg_Y A < q$ and every coefficient in X , as well as B , has degree at most E . The polynomial A has poles only at infinity, with total pole degree at most $q(E + N)$; the same bound holds for B . The degree of its zero divisor equals its pole degree, so no nonzero numerator or denominator can have an order of vanishing greater than this bound at a rational point. Consequently local substitutions through precision

$$8q(E + N + 1) \tag{5.4}$$

suffice to identify their first nonzero Laurent coefficients, even after the possible pole at infinity has been removed. For a represented divisor, its coefficient at a specified finite point is the minimum of the valuations of a list of ideal generators; zero generators may be discarded.

We next turn Riemann–Roch existence into a linear computation. Recall that

$$L(V) = \{a \in \mathcal{L} : \operatorname{div}(a) + V \geq 0\} \cup \{0\}$$

is a finite-dimensional k -vector space. Given D and $\ell \geq 0$, choose a bound H for $I(-D)$. Every function in $L(D + \ell\infty_0)$ has the form

$$a = H^{-1} \sum_{j=0}^{q-1} a_j(X) Y^j, \quad \deg a_j \leq \deg H + \|D\| + \ell. \tag{5.5}$$

To justify the degree bound, put

$$M = \max_{a_j \neq 0} (\deg a_j + jN/q - \deg H).$$

The coefficient of z^{-M} in the expansions at infinity is a nonzero polynomial of degree less than q evaluated at the q distinct values $1, \zeta, \dots, \zeta^{q-1}$. It cannot vanish at all of them. Thus at one infinity point the pole order, interpreted as $-v_{\infty_j}(a)$, is exactly M . The defining inequalities for $L(D + \ell\infty_0)$ give $M \leq \|D\| + \ell$, which implies (5.5).

In this finite ansatz, membership in $I(-D)$ is a subspace condition, and

$$v_{\infty_j}(a) \geq -v_{\infty_j}(D) - \ell \mathbf{1}_{j=0} \quad (0 \leq j < q)$$

is a list of linear conditions on Laurent coefficients. Expansions to the bound (5.4), increased by $8q(\|D\| + \ell + 1)$, suffice. Solving the resulting linear system therefore computes the whole space $L(D + \ell\infty_0)$.

For degree-zero D , the Riemann–Roch inequality [29, Tag 0BS6] gives $\dim_k L(D + g\infty_0) \geq 1$. Choose a nonzero solution a by a fixed pivot convention and define

$$\operatorname{ReduceDivisor}(D) = D' = D + \operatorname{div}(a). \tag{5.6}$$

Since $D' + g\infty_0$ is effective of degree g , we have $\|D'\| \leq 2g$. Retain a whenever a later computation needs the linear equivalence. If a degree-zero divisor D is known to be principal, taking $\ell = 0$ instead produces a nonzero a with $D + \text{div}(a) = 0$: the effective divisor on the left has degree zero.

All vector spaces and polynomial degrees in these constructions are polynomial in the parameters in Proposition 5.1. Dense polynomial operations, fraction reduction over $k(X)$, and linear algebra therefore prove its complexity assertion. For example, the ideal quotient has dimension $2q \deg H$, and the ansatz (5.5) has at most $q(\deg H + \|D\| + \ell + 1)$ unknowns. Rational linear algebra does not cause uncontrolled degree growth: after common denominators are cleared, minors of an $s \times s$ matrix with polynomial entries of degree at most d have degree at most sd . Reduced elimination entries and solutions are ratios of such minors. This completes the proof of Proposition 5.1.

5.3 Summing divisors given over a product algebra

The odd-degree splitter first produces one divisor for each root of F , through a computation over $A = k[T]/F(T)$. The next lemma computes their sum over k , while the roots remain unknown. A norm from the finite algebra A will replace the unavailable product over its components.

Lemma 5.3 (Simultaneous divisor sum). *Suppose $A \simeq k^s$ is an explicitly represented split algebra. Let $I_1, \dots, I_s$ be fractional ideals of O supplied together over A , with monic bounds H_i and a common number b of generators $u_{i0}, \dots, u_{i,b-1}$, including H_i . If the prime subfield contains more than $s(b-1)$ elements, their product can be computed by polynomial arithmetic, determinants, and ideal closure, without a decomposition of A . Together with addition of the infinity vectors, this computes the sum of the corresponding divisors.*

Proof. Let $H \in A[X]$ denote the tuple of bounds and let u_j denote the tuple of j th generators. Form

$$H_* = \text{Norm}_{A/k}(H) = \prod_{i=1}^s H_i, \quad w_a = \text{Norm}_{A/k} \left(\sum_{j=0}^{b-1} a^j u_j \right) = \prod_{i=1}^s \left(\sum_{j=0}^{b-1} a^j u_{ij} \right).$$

The norm is the determinant of multiplication after extending scalars to the relevant polynomial or function ring. The displayed products are identities proving correctness, not instructions to recover the components. Denominators are cleared by multiplying the i th component by H_i ; the resulting determinant lies in O , and division by H_* recovers w_a . Polynomial determinants can be computed before reducing by $Y^q - F(X)$. Use $s(b-1) + 1$ distinct values of a from the prime subfield.

Each w_a belongs to $\prod_i I_i$. Fix a finite closed point P and write $d_i = \min_j v_P(u_{ij})$. After dividing the i th combination by a local uniformizer to power d_i , its residue is a nonzero polynomial in a , over $k(P)$, of degree at most $b-1$. At most $b-1$ of our values fail to attain valuation d_i . Thus some chosen a attains all d_i simultaneously, and $v_P(w_a) = \sum_i d_i$. Fractional ideals in a Dedekind domain are determined by these minimum valuations. The w_a therefore generate exactly $\prod_i I_i$. The polynomial H_* is a valid bound, so finite ideal closure computes its stored representation.

Infinity coefficients are integers. In a simultaneous field execution they are the same in every surviving component: any different outcome of a scalar zero test would already have supplied a factor. Their sum is therefore obtained by multiplying the common vector by s . More generally, if the computation has retained separate pieces, sum their vectors with the corresponding dimensions. $\square$

For the application $s = N$. The number b can be bounded by $2q \deg H_i + 1$ in the compressed representation, and the characteristic cutoff in Section 10 guarantees the required distinct scalars. Matrix determinants and ideal closure have sizes polynomial in these parameters.

5.4 Binary coefficients at infinity and function circuits

Riemann–Roch reduction has cost polynomial in absolute degree. It must not be applied directly to an infinity divisor whose coefficients are exponentially large integers. We instead perform the class arithmetic by binary additions and retain the accompanying rational function as a circuit.

Lemma 5.4 (Principal functions with large infinity coefficients). *Let D be a represented degree-zero divisor of bounded absolute degree, and let $W = \sum_j w_j \infty_j$ have degree zero, with integer coefficients given in binary. If $Z = (1 - \sigma)D - W$ is principal, a circuit for a function h satisfying $\text{div}(h) = Z$ can be computed using a number of operations polynomial in $q, N, \|D\|$, the representation size, and $\max_j \log(2 + |w_j|)$. The circuit uses multiplication and inversion of explicitly represented small functions. At each known ramification point P , it computes the nonzero value $h(P)$ within the same type of bound, without expanding h .*

Proof. Write $W = \sum_{j=1}^{q-1} w_j(\infty_j - \infty_0)$. Build the divisor Z from $(1 - \sigma)D$ and these signed multiples by binary doubling and addition. At every stage retain a reduced divisor D_0 and a function circuit w with the invariant

$$D_0 = D_{\text{target}} + \text{div}(w). \quad (5.7)$$

On adding two stages, reduce $D_1 + D_2$ with reducing function u and replace the function circuit by $w_1 w_2 u$. For doubling it becomes $w_1^2 u$; negation uses inversion followed, if needed, by reduction. After the initial reduction, each materialized sum has absolute degree at most $4g$, and the number of reductions is $O(q \max_j \log(2 + |w_j|))$, in addition to the initial operations on D .

At the end obtain a small divisor $\tilde{Z} = Z + \text{div}(w)$. It is principal, so the $\ell = 0$ Riemann–Roch computation gives a with $\tilde{Z} + \text{div}(a) = 0$. Consequently

$$h = (aw)^{-1}$$

has divisor Z . All leaves of this circuit are functions produced by reductions on small divisors and hence have polynomial representation size.

At P , expand each leaf and store its integer valuation and its nonzero leading coefficient. Multiplication adds valuations and multiplies leading coefficients; inversion negates valuations and inverts leading coefficients. The resulting integers have bit length polynomial in the circuit length and leaf bounds. Because P is fixed by σ and W is supported at infinity, $v_P(Z) = 0$. The final valuation is therefore zero, and the final leading coefficient is exactly $h(P)$. This also explains why the intermediate objects are leading coefficients of Laurent series, rather than values of functions that may have poles. $\square$

6 Solving the promised norm equations

Division of divisor classes will require a function with a prescribed norm. This section constructs such a function when its existence is known. The construction uses linear algebra and extraction of roots in finite reduced algebras, and its complexity is polynomial in the degree of the cover. In particular, the cover degree is not treated as a constant.

Theorem 6.1 (Promised norm solver). *Let k be an explicitly represented finite field of characteristic p , let q be an odd prime, and suppose that a primitive q th root of unity $\zeta \in k$ and a generator of the full q -primary subgroup of k^* are given. Let $F \in k[X]$ be monic and square-free, of positive degree N divisible by q . Put*

$$K_0 = k(X), \quad \mathcal{L} = K_0[Y]/(Y^q - F), \quad \rho(Y) = \zeta Y.$$

Let $G \in K_0^$, write D for the maximum of its reduced numerator and denominator degrees, and assume*

$$p > (N + 2D + 1)^2. \quad (6.1)$$

If G belongs to $\text{Norm}_{\mathcal{L}/K_0}(\mathcal{L}^)$, a deterministic algorithm constructs $h \in \mathcal{L}^*$ with*

$$\prod_{j=0}^{q-1} \rho^j(h) = G.$$

Its bit complexity, and the degrees of the rational coefficients of h in the basis $1, Y, \dots, Y^{q-1}$, are bounded by fixed polynomials in q, N, D , and $\log |k|$. It uses no polynomial factorization oracle. Its field-dependent branches use only zero tests, so the algorithm admits the simultaneous execution of Lemma 2.1.

The characteristic bound is a convenient sufficient condition for the scalar searches in Lemma 2.2; the main algorithm satisfies it with considerable room. The roots of F need not be supplied, and the norm solver does not require F to split over k . The function automorphism ρ is the inverse of the action σ used in the divisor sections. Their powers run over the same cyclic group, so they define the same norm.

We first turn the norm equation into the problem of constructing a rank-one idempotent in an algebra. We then obtain that idempotent from spaces of matrices regular at the places of the rational function field. This use of intersections of maximal orders is closely related to the explicit matrix-algebra algorithms of Ivanyos–Kutas–Rónyai [14]. Their general algorithm permits a finite-field factorization oracle. Here the cyclic presentation allows the local orders to be written explicitly, and a fiber calculation supplies the idempotent by linear algebra. We give the construction and its bounds in full.

6.1 The cyclic algebra and its local matrix models

The polynomial $Y^q - F$ is irreducible over K_0 : a place at which F has valuation one is Eisenstein for it. Thus $\mathcal{L}$ is a field, cyclic of degree q over K_0 . Form the K_0 -algebra

$$\mathcal{E} = \bigoplus_{b=0}^{q-1} \mathcal{L}V^b, \quad Va = \rho(a)V \ (a \in \mathcal{L}), \quad V^q = G. \quad (6.2)$$

Its multiplication table in the basis

$$W_{ab} = Y^a V^b, \quad 0 \leq a, b < q,$$

is explicit. The norm promise implies that $\mathcal{E}$ is a full matrix algebra over K_0 , but does not initially give us an isomorphism. Indeed, if h_0 is any norm solution, let $\mathcal{L}$ act on itself by multiplication and let V act by $h_0\rho$. These operators satisfy (6.2). Their linear independence follows by applying a putative relation between the operators ρ^b to $1, Y, \dots, Y^{q-1}$ and using the invertible matrix $(\zeta^{bj})_{b,j}$. Dimension then gives an isomorphism

$$\mathcal{E} \simeq M_q(K_0). \quad (6.3)$$

This argument establishes existence only; the algorithm never uses h_0 or this unspecified isomorphism.

A *rank-one idempotent* in $\mathcal{E}$ is an element e with $e^2 = e$ whose image under (6.3) has rank one. The following observation explains why constructing such an element solves the original norm equation.

Lemma 6.2. *Given a rank-one idempotent $e \in \mathcal{E}$, linear algebra over K_0 constructs a norm solution $h \in \mathcal{L}$.*

Proof. The left ideal $\mathcal{E}e$ has dimension q over K_0 . The map $\mathcal{L} \rightarrow \mathcal{E}e$, $a \mapsto ae$, is injective, since any nonzero $a \in \mathcal{L}$ is invertible. Both sides have dimension q , so it is an isomorphism. In particular, there is a unique $h \in \mathcal{L}$ satisfying

$$Ve = he. \quad (6.4)$$

Solve this equation for the q coefficients of h in the basis $1, Y, \dots, Y^{q-1}$, using the explicit q^2 -element basis of $\mathcal{E}$ to express both sides. Repeatedly multiply the equation on the left by V , using $Va = \rho(a)V$. After q steps this gives

$$Ge = V^q e = \left(\prod_{j=0}^{q-1} \rho^j(h) \right) e.$$

Since $e \neq 0$ and the two coefficients are scalars in K_0 , their equality proves the norm equation. In particular $h \neq 0$. $\square$

Our remaining task is to construct e . We will impose local integrality conditions on elements of $\mathcal{E}$, then find a rank-one matrix in their values at infinity.

Let v be a closed point of $\mathbb{P}_k^1$. Write K_v for the completed rational function field, R_v for its valuation ring, κ_v for its residue field, and T for a uniformizer. A *lattice* in a finite-dimensional K_v -space is a free R_v -module of full rank. An *order* in $\mathcal{E} \otimes K_v$ is a lattice that contains 1 and is closed under multiplication; it is *maximal* if no strictly larger order contains it. We construct one maximal order $\mathcal{A}_v$ at every v . Only the points dividing F , the numerator or denominator of G , and the point at infinity require nontrivial conditions.

We will repeatedly use the matrix trace tr on the split algebra. It does not depend on the choice of isomorphism in (6.3). Conjugation by Y and V shows that every nonidentity basis monomial has trace zero, while $\text{tr}(1) = q$. Consequently

$$a = \sum_{a', b'} c_{a'b'} W_{a'b'} \implies c_{ab} = q^{-1} \text{tr}(a W_{ab}^{-1}). \quad (6.5)$$

Here q is invertible by (6.1).

Unramified places with $q \mid v(G)$. Suppose v is finite, $v(F) = 0$, and $w = v(G)$ is divisible by q . Set $\tilde{V} = T^{-w/q}V$, and take the R_v -span of $Y^a \tilde{V}^b$. Both Y and $\tilde{V}$ have unit q th powers, so this span is an order and contains the inverses of all its basis monomials. It is maximal. To see this, any larger order has integral matrix traces: its regular left-action trace is integral and equals q times matrix trace. Pairing an element of the larger order with the inverse basis monomials in (6.5) therefore makes all its coordinates integral. It already belongs to the displayed span. This description applies at all finite places outside the indicated finite set, with $w = 0$.

Places ramified in $\mathcal{L}$. Suppose $v(F) = 1$, and again put $w = v(G)$. Since q is odd,

$$V' = Y^{-w}V, \quad (V')^q = u := G/F^w \in R_v^*. \quad (6.6)$$

The commutation factor in taking the q th power is a power of $\zeta^{q(q-1)/2} = 1$. The residue of u is a q th power in κ_v^* . Indeed the extension at v is totally ramified of degree q . For any promised solution h_0 one has $v_{\mathcal{L}}(h_0) = v(G) = w$, with $v_{\mathcal{L}}(Y) = 1$. Hence h_0/Y^w is a unit, its norm is u , and the residue of that norm is the q th power of its residue.

Take a residue root and lift it by Hensel's method to $R \in R_v^*$ with $R^q = u$. On column vectors $e_0, \dots, e_{q-1}$ define

$$V'e_j = \zeta^j R e_j, \quad Y e_j = \begin{cases} e_{j+1}, & j < q-1, \\ F e_0, & j = q-1. \end{cases} \quad (6.7)$$

These matrices satisfy $V'Y = \zeta Y V'$, $Y^q = F$, and $(V')^q = u$. The distinct diagonal entries of V' give all diagonal matrix units by interpolation, and the invertible shift Y gives the remaining matrix units over K_v . Thus the model is an isomorphism to $M_q(K_v)$. Let $\mathcal{A}_v$ be the inverse image of $M_q(R_v)$.

Unramified places with $q \nmid v(G)$. Suppose $v(F) = 0$ and $q \nmid w = v(G)$. The residue of F must be a q th power. Otherwise, since κ_v contains μ_q and q is prime, $Y^q - \bar{F}$ is irreducible and defines an unramified extension of degree q . All its norm valuations are then divisible by q , contradicting the promise on G . Lift a residue root to $S \in R_v^*$ with $S^q = F$. Use the matrices

$$Y e_j = \zeta^{-j} S e_j, \quad V e_j = \begin{cases} e_{j+1}, & j < q-1, \\ G e_0, & j = q-1. \end{cases} \quad (6.8)$$

Now $VY = \zeta Y V$, including at the wrap from e_{q-1} to e_0 . The same matrix-unit argument gives an isomorphism to $M_q(K_v)$; again take the inverse image of $M_q(R_v)$.

Infinity. Put $z = X^{-1}$ and $\tilde{Y} = z^{N/q}Y$. Then

$$\tilde{Y}^q = z^N F(z^{-1}) =: F_{\infty}(z), \quad F_{\infty}(0) = 1.$$

There is a unique root $S \in k[[z]]$ of $S^q = F_{\infty}$ with $S(0) = 1$. Use (6.8) with $\tilde{Y}$ in place of Y , and with the same weighted shift for V , whatever the valuation of G may be. The inverse image of $M_q(k[[z]])$ is $\mathcal{A}_{\infty}$. This model also specifies reduction to $M_q(k)$ at infinity.

The matrix orders just used are maximal. For example, pairing with matrix units shows that any element in a larger order has integral matrix entries, using the integrality of matrix trace exactly as above. We have therefore specified maximal orders at every place, using only residue roots whose existence follows from the norm promise.

6.2 Why one fiber contains a rank-one matrix

Define the finite-dimensional k -algebra of global sections and its evaluation image by

$$\mathcal{S} = \{a \in \mathcal{E} : a \in \mathcal{A}_v \text{ for every } v\}, \quad \mathcal{B} = \text{ev}_{\infty}(\mathcal{S}) \subseteq M_q(k). \quad (6.9)$$

We will compute both spaces by linear equations. First we establish the feature of $\mathcal{B}$ that will make this useful: up to change of basis it is the algebra of all block upper triangular matrices. Its first block therefore contains the image of a rank-one idempotent.

Here is the geometric explanation, including the descent from completed local orders. Intersect $\mathcal{A}_v$ with the rational algebra $\mathcal{E}$. This is a full lattice over the ordinary local discrete valuation ring at v , and its completion is $\mathcal{A}_v$. Indeed the completed lattice lies between two powers of a rational uniformizer times a fixed rational lattice. Rational approximation to a basis, to sufficiently high precision, produces a basis of the same completed lattice. Equivalently, the finitely many congruences between these two bounds descend to the ordinary local ring. Maximality descends as well, since a larger ordinary order would yield a larger completed order.

For the following existence argument only, fix an isomorphism (6.3). An ordinary local maximal order in $M_q(K_0)$ is the endomorphism ring of a lattice. In fact, apply the order to the standard local lattice R^q and take its span M . The order is a finite module and contains the identity, so M is a full lattice stable under it. The containing order $\text{End}_R(M)$ must equal it by maximality. At all but finitely many places the standard rational basis matrices and their inverse change of coordinates are integral; there we may use $M = R^q$. The resulting lattices glue to a vector bundle $\mathcal{V}$ of rank q on $\mathbb{P}_k^1$. Concretely, its local sections are rational column vectors satisfying the specified lattice conditions. A basis of one local lattice gives a trivialization on a neighborhood after excluding the finitely many poles and other exceptional points. We have proved

$$\mathcal{S} = H^0(\mathbb{P}_k^1, \text{End } \mathcal{V}). \quad (6.10)$$

The chosen matrix model at infinity identifies the endomorphism fiber with $M_q(k)$. Any two such identifications differ by conjugation: an automorphism of a full matrix algebra sends matrix units to matrix units, from which a change of column basis is obtained. Thus its particular choice will not affect the asserted shape of $\mathcal{B}$.

The following form of the projective-line splitting theorem holds over the original field, rather than only after extending its constants; see Hazewinkel–Martin [12].

Lemma 6.3 (Splitting on the projective line). *Every vector bundle on $\mathbb{P}_k^1$ is a direct sum of line bundles $\mathcal{O}(d)$, with integer degrees d .*

Proof. We recall a proof over the given field k . Line bundles on $\mathbb{P}_k^1$ are classified by their degree. On the two standard affine charts, the Laurent-polynomial computation of their cohomology gives

$$H^0(\mathcal{O}(t)) = 0 \ (t < 0), \quad H^1(\mathcal{O}(t)) = 0 \ (t \geq -1).$$

The first assertion is immediate from polynomial degrees; for the second, the Čech quotient is spanned by the Laurent monomials between the two chart ranges, and that range is empty for $t \geq -1$.

Choose the largest integer d for which there is a nonzero map $\mathcal{O}(d) \rightarrow \mathcal{V}$. Such integers exist and are bounded above: enclosing the finitely many defining lattices of $\mathcal{V}$ between twists of a fixed trivial bundle proves both claims. A nonzero map is generically injective. Saturating its image would increase its degree by the effective divisor of its zeros, so maximality of d implies that it is a line subbundle. The quotient $\mathcal{Q}$ is a vector bundle, and induction on rank writes it as $\bigoplus_i \mathcal{O}(t_i)$. Twist

$$0 \longrightarrow \mathcal{O}(d) \longrightarrow \mathcal{V} \longrightarrow \mathcal{Q} \longrightarrow 0$$

by $-d-1$. Maximality of d gives $H^0(\mathcal{V}(-d-1)) = 0$, and $H^1(\mathcal{O}(-1)) = 0$. The cohomology sequence therefore gives $H^0(\mathcal{Q}(-d-1)) = 0$, whence $t_i \leq d$ for every i . Finally, each extension group

$$\text{Ext}^1(\mathcal{O}(t_i), \mathcal{O}(d)) = H^1(\mathcal{O}(d-t_i))$$

vanishes. The sequence splits, completing the induction. $\square$

$$\mathcal{B} \sim \begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{pmatrix} \xrightarrow{\ker \operatorname{tr}(ab)} \mathcal{N} \sim \begin{pmatrix} 0 & * & * \\ 0 & 0 & * \\ 0 & 0 & 0 \end{pmatrix}, \quad U \sim \begin{pmatrix} k^{r_1} \\ 0 \\ 0 \end{pmatrix}.$$

Figure 3: The fiber calculation with three distinct degree blocks. Each star denotes an arbitrary matrix block, and the diagonal block sizes are r_1, r_2, r_3 . The common kernel U of the trace-pairing kernel is the first block column space. Every matrix vw with $v \in U$ belongs to $\mathcal{B}$; choosing $wv = 1$ gives the rank-one idempotent. The calculation works in the original coordinates.

Apply the lemma and order the summands by decreasing degree: $\mathcal{V} = \bigoplus_{i=1}^q \mathcal{O}(d_i)$ with $d_1 \geq \dots \geq d_q$. The (i, j) entry of a global endomorphism is a section of $\mathcal{O}(d_i - d_j)$. If $d_i < d_j$ it is zero; otherwise its value at infinity is arbitrary, because a line bundle of nonnegative degree has a section with any prescribed value at that point. Grouping the equal degrees proves the promised block upper triangular description of $\mathcal{B}$. This is an existence proof of the shape, not an algorithm for computing the splitting of $\mathcal{V}$.

Lemma 6.4 (Extracting a rank-one fiber idempotent). *Given a basis of a subalgebra $\mathcal{B} \subseteq M_q(k)$ that is conjugate to a full block upper triangular algebra, linear algebra over k constructs a rank-one idempotent in $\mathcal{B}$.*

Proof. Compute the kernel $\mathcal{N}$ of the bilinear form

$$(a, b) \longmapsto \operatorname{tr}(ab), \quad a, b \in \mathcal{B}.$$

In block upper triangular coordinates, a product has diagonal blocks equal to the products of the corresponding diagonal blocks. The ordinary trace pairing on a full matrix algebra is nondegenerate in every characteristic, as is seen by pairing its matrix units. Hence $\mathcal{N}$ is exactly the strictly block upper triangular part. The common kernel

$$U = \bigcap_{n \in \mathcal{N}} \ker n$$

is the column space of the first block; if there is only one block, $\mathcal{N} = 0$ and $U = k^q$. Indeed every later block has arbitrary maps to the first block, which exclude any vector with a nonzero later component from this common kernel.

Choose $0 \neq v \in U$ and a row w with $wv = 1$. Then $a_0 = vw$ has rank one and satisfies $a_0^2 = a_0$. In block coordinates it has nonzero entries only in the first block row, so it belongs to $\mathcal{B}$. The computation requires only kernels and one scalar normalization; it does not require finding the conjugating matrix. $\square$

Figure 3 isolates the part of the vector-bundle argument that is actually computed. The splitting theorem guarantees the triangular shape; the trace pairing finds enough of that shape to construct a rank-one matrix without a factorization or a change of basis.

6.3 From the fiber idempotent to a norm solution

Lift the matrix a_0 of Lemma 6.4 to any $b \in \mathcal{S}$ with $\operatorname{ev}_\infty(b) = a_0$. This is a linear solve once (6.9) has been computed. The coefficients of the generic characteristic polynomial of b are regular at every point, since b is a global endomorphism of $\mathcal{V}$. A rational function regular on $\mathbb{P}_k^1$ is constant. Evaluating at infinity consequently gives

$$\det(T - b) = T^{q-1}(T - 1).$$

The primary decomposition for the relatively prime factors T^{q-1} and $T - 1$ shows that

$$e = b^q \tag{6.11}$$

is the projection onto the one-dimensional 1-eigenspace: it is zero on the nilpotent part and the identity on that eigenspace. Thus e is a rank-one idempotent in $\mathcal{E}$.

Lemma 6.2 now recovers a norm solution by solving $Ve = he$ over K_0 .

It remains to make the spaces in (6.9) computable with polynomially many coefficients. This is where the explicit local models, rather than an unspecified splitting of the cyclic algebra, are essential.

6.4 Computing all sections by finite linear systems

Write $G = A/B$ with coprime nonzero polynomials $A, B \in k[X]$. Square-free decomposition and gcds partition the finite places dividing FAB into pairwise coprime monic square-free polynomials H_i such that the two values

$$(v(F), v(G)) = (\epsilon_i, w_i)$$

are constant over the irreducible factors of each H_i . No irreducible factorization is involved: decompose the multiplicity layers of A and B and refine their intersections with each other and with F by gcds. All these polynomials have degree less than p by (6.1), so derivative-based square-free decomposition has no inseparable residual case. Put

$$H = \prod_i H_i, \quad d_H = \deg H \leq N + 2D, \quad M = 10q^2(N + D + 1). \tag{6.12}$$

At every place in group i we may use the same rational uniformizer $T = H_i$. Arithmetic modulo H_i^s carries out, simultaneously, all the required arithmetic modulo the s th power of these local uniformizers. The Chinese remainder theorem explains this statement without requiring the factors of H_i to be computed. When a local model requires a residue root, its residue is a unit modulo H_i and is a q th power in every residue-field component by the norm promise. Lemma 2.2 constructs the root in $k[X]/H_i$. The characteristic bound supplies more than d_H^2 distinct prime field scalars for that lemma. Hensel lifting then works modulo powers of H_i , since the derivative qR^{q-1} or qS^{q-1} is a unit.

Lemma 6.5 (Bounds for the local models). *At a place with one of the matrix models above, every nonzero entry of the matrices of W_{ab} and W_{ab}^{-1} has valuation between $-M$ and M . At a place with the scaled-basis order, the exponents changing between that basis and W_{ab} have absolute value at most M . Consequently, the standard coefficients of every local-order element have valuation at least $-M$ at the exceptional places and at infinity. At all other finite places those coefficients are integral.*

Proof. At a ramified finite place, let D_R denote the unit diagonal matrix of V' . Then $V = Y^w D_R$, and commutation gives

$$W_{ab} = \zeta^{wb(b-1)/2} Y^{a+wb} D_R^b.$$

For any integer t , the nonzero entry of Y^t in column j has valuation $\lfloor (j+t)/q \rfloor$. Since $0 \leq a, b < q$ and $|w| \leq D$, these valuations, and those of the inverse matrices, have absolute value at most $D + 2$. In the unramified matrix model, Y is a unit diagonal matrix and each entry of V^b , for $b < q$, has valuation either 0 or w ; inverse entries have the opposite valuations. At infinity the extra factor $z^{-aN/q}$ in Y^a gives the bound $N + D$. These bounds

are smaller than M . In the scaled-basis case, the change is multiplication by $T^{\pm bw/q}$, whose exponent has absolute value at most D .

For an element in a matrix order, formula (6.5) and the bound on the inverse basis matrices give the coefficient bound $-M$. The scaled-basis assertion gives the same conclusion in the remaining case. At an ordinary finite place the order is the unscaled standard span, so the coefficients are integral there. $\square$

It follows that every global section is included in the ansatz

$$b = \sum_{a,b'=0}^{q-1} c_{ab'} W_{ab'}, \quad c_{ab'} = \frac{P_{ab'}(X)}{H(X)^M}, \quad \deg P_{ab'} \leq M(d_H + 1). \quad (6.13)$$

Indeed the denominator clears every finite pole, and the bound at infinity says precisely that its numerator degree exceeds its denominator degree by at most M . The polynomial coefficients in (6.13) are our scalar unknowns.

For each finite group impose membership in $\mathcal{A}_v$ as follows. In a matrix model compute, modulo T^{2M+1} , the integral quantities $T^M c_{ab'}$ and every matrix entry of $T^M W_{ab'}$. Their products and sums give T^{2M} times the matrix of the section. The section is integral exactly when all coefficients below order $2M$ vanish. These are linear conditions over k . In a scaled-basis order use the coordinates in that basis in place of matrix entries and impose the identical divisibility condition. At infinity make the same computation with $T = z$ and the prescribed matrix model. The coefficient of z^{2M} , after the lower ones vanish, is exactly the evaluated matrix in $M_q(k)$.

For completeness, the truncated entries are computed without an infinite-series oracle. Each nonzero basis entry has the form $T^e u$ for a known exponent e with $|e| \leq M$ and a unit u given by the diagonal and weighted-shift formulas. Lift the required unit roots to precision $3M + 3$ by the simple-root Hensel iteration. After multiplication by T^M , this precision more than supplies all terms through T^{2M} . For the rational coefficients of (6.13), cancelling T^M from the denominator leaves a unit denominator at the group, inverted by the polynomial Euclidean algorithm. At infinity replace X by z^{-1} and cancel the known powers of z . The same finite-precision computation then applies.

Solving these linear equations gives a basis of $\mathcal{S}$ together with the evaluated matrix of every basis vector. Taking their span gives $\mathcal{B}$ and retains a linear map with which to lift its elements. The ansatz contains every section by Lemma 6.5; the imposed conditions are necessary and sufficient at every exceptional place, and elsewhere the ansatz is already integral. Thus the computed spaces are exactly those in (6.9).

6.5 Algorithm and polynomial bounds

The complete procedure can now be stated without any existential computational step.

1. Form the multiplication table of $\mathcal{E}$ in the basis $Y^a V^b$. Compute the square-free groups H_i and the local models, obtaining their residue roots by Lemma 2.2 and their needed precision by Hensel lifting.
2. Use (6.13) and the divisibility conditions above to compute $\mathcal{S}$ and its evaluation map to $M_q(k)$.
3. In the image $\mathcal{B}$, compute the trace-pairing kernel, its common column kernel, and a rank-one idempotent $a_0 = vw$ as in Lemma 6.4.
4. Lift a_0 to $b \in \mathcal{S}$ by a linear solve and compute $e = b^q$ with the multiplication table.

5. Solve $Ve = he$ for $h \in \mathcal{L}$ over $k(X)$ and return h .

Every step is deterministic after fixing a basis order and the first available pivot in each linear solve. The preceding arguments prove existence of the required roots, a nonzero vector in the common kernel, a lift of a_0 , and a solution of the last system; therefore the procedure terminates on every promised input.

We record bounds to ensure that this is also a uniform algorithm. There are at most d_H groups, and

$$d_H \leq N + 2D, \quad M = 10q^2(N + D + 1).$$

The number of unknowns in (6.13) is $q^2(M(d_H + 1) + 1)$. The total number of scalar local conditions is at most a constant times $q^2M(d_H + 1)$, since precision at a group of degree d_i contributes $O(Md_i)$ coefficients. All quotient polynomial rings have dimension at most $(3M + 3)d_H$ over k . These are fixed polynomial bounds in q, N, D . The finite-field root calculations also have polynomial complexity by Lemma 2.2; the bit lengths of their powering exponents are $O(d_H \log |k|)$.

For explicit denominator control, compute b^q by successive multiplications by b . Write $P_* = M(d_H + 1)$. The coefficients of b^s have the common denominator

$$H^{Ms} B^{s-1}$$

and numerator degrees at most $sP_* + (s - 1)(N + D)$. Indeed a product of two standard basis monomials introduces at most one factor of F and one factor of $G = A/B$. Passing to the displayed common denominator at the next multiplication therefore adds at most $P_* + N + D$ to numerator degree, including every addition of terms. This proves the assertion by induction for $1 \leq s \leq q$. The final system $Ve = he$ has the same common denominator with at most one extra factor of B ; its cleared entries have polynomial degree. A solution of this consistent rank- q system is expressed by ratios of minors, of degrees at most q times the cleared entry degree. Fraction-reducing elimination or these minor formulas therefore bounds the output degree and the number of field operations by fixed polynomials. Standard dense polynomial arithmetic and Gaussian elimination convert these bounds to a fixed polynomial in $q, N, D, \log |k|$ bit operations. The more generous common bound needed by the full factorization algorithm is recorded in Proposition 10.1.

Finally, polynomial gcds, Hensel lifting, and linear elimination use only field arithmetic and zero tests. Integer choices such as degrees and multiplicities are determined by those tests; the valuation groups are formed by polynomial gcds rather than by an unknown factorization. The finite-field root routine has the same property. Hence all steps are compatible with simultaneous execution, proving the last assertion of Theorem 6.1 and completing its proof.

7 Effective division by $1 - \sigma$

The geometric argument supplies divisor classes that are divisible by $\lambda = 1 - \sigma$ over the working field. We now construct a divisor representing such a division. Its only nonlinear instruction is the promised norm equation solved in Theorem 6.1. A ramification point fixes the otherwise undetermined constant in that equation.

Proposition 7.1 (Class division). *Let k, q, F, C , and σ be as in Proposition 5.1, and suppose the auxiliary primary-group data required by Theorem 6.1 are available. Let $P = (x, 0)$ be a specified k -rational ramification point. Given a represented degree-zero divisor D satisfying $\text{char } k > (N + 2\|D\| + 1)^2$, under the promise that*

$$[D] \in (1 - \sigma) \text{Pic}^0(C)(k),$$

there is a deterministic procedure $\text{LambdaDivide}(D, P)$ returning a degree-zero divisor D' with

$$\|D'\| \leq 2g, \quad (1 - \sigma)[D'] = [D].$$

It uses polynomially many field operations in the representation size, $q, N, \|D\|$, and $\log |k|$, with the uniform norm bound from Theorem 6.1. It is compatible with simultaneous execution over the algebra of the unknown roots of F .

The procedure consists of three steps.

1. Compute $j = h^+/h^-$ by Lemma 5.2. Put $b = v_{X=x}(j)$ and normalize $G = cj$ so that G/F^b has residue 1 at $X = x$.
2. Use Theorem 6.1 to find a with $\text{Norm}(a) = G$. Set $E = D - \text{div}(a)$, and form the coefficientwise maximum V of $0, E, E + \sigma E, \dots, E + \dots + \sigma^{q-2}E$.
3. Return $\text{ReduceDivisor}(V - (\deg V)P)$.

We prove first that the norm equation in the second step is solvable, and then that the maximum gives the required class division.

Proof. Lemma 5.2 gives $j = h^+/h^- \in k(X)^*$ with

$$\text{div}_C(j) = \sum_{a=0}^{q-1} \sigma^a D. \quad (7.1)$$

Put $b = v_{X=x}(j)$. Since F has a simple root at x , the rational function j/F^b is a unit there, even when b is negative. Compute its nonzero residue and set

$$c = \left(\frac{j}{F^b} \Big|_{X=x} \right)^{-1}, \quad G = cj. \quad (7.2)$$

Thus G/F^b has residue 1 at x .

We first justify that G is a norm from $\mathcal{L} = k(C)$ to $k(X)$, which is the promise needed by the norm algorithm. Choose a rational divisor D_0 representing a class with $(1 - \sigma)[D_0] = [D]$. Lemma 4.4 supplies rational representatives and rational functions for rational principal divisors. There is therefore an $a_0 \in \mathcal{L}^*$ with

$$\text{div}(a_0) = D - (1 - \sigma)D_0.$$

Taking orbit sums and using (7.1) shows

$$\text{Norm}_{\mathcal{L}/k(X)}(a_0) = c_0 j \quad \text{for some } c_0 \in k^*.$$

At P the cover is totally ramified, with residue field k , so the norm valuation formula gives $v_P(a_0) = b$. Also

$$\text{Norm}_{\mathcal{L}/k(X)}(Y) = F$$

because q is odd. The function $u = a_0/Y^b$ is thus a unit at P and has norm c_0j/F^b . Every deck transformation acts trivially on its residue, whence

$$\left. \frac{c_0j}{F^b} \right|_{X=x} = u(P)^q.$$

By (7.2), the left side is c_0/c . It follows that c/c_0 is a q th power in k^* , and hence that $G = cj$ is a norm. This argument is solely a proof of the promise; neither D_0 nor a_0 is an input to the algorithm. Moreover, Lemma 5.2 bounds the numerator and denominator degrees of G by $\|D\|$, so the stated characteristic hypothesis implies the one in Theorem 6.1.

Apply Theorem 6.1 to obtain $a \in \mathcal{L}^*$ with $\text{Norm}(a) = G$, and put

$$E = D - \text{div}(a).$$

Equation (7.1) gives $\sum_{a=0}^{q-1} \sigma^a E = 0$. Define the partial sums and their coefficientwise maximum by

$$S_0 = 0, \quad S_j = \sum_{a=0}^{j-1} \sigma^a E \quad (1 \leq j \leq q), \quad V = \max_{0 \leq j < q} S_j. \quad (7.3)$$

Since $S_q = S_0 = 0$ and $\sigma S_j = S_{j+1} - E$, cyclically indexing the same list shows

$$\sigma V = V - E, \quad (1 - \sigma)V = E.$$

The maximum is effective because the list includes 0. As P is fixed by σ , the divisor $V - (\deg V)P$ has degree zero and satisfies

$$(1 - \sigma)[V - (\deg V)P] = [E] = [D].$$

Return $\text{ReduceDivisor}(V - (\deg V)P)$. This proves the asserted class identity and absolute-degree bound.

For completeness, every instruction in this description is effective in the required representation. The valuation and residue in (7.2) are obtained by polynomial division at $X - x$. The numerator and denominator of j have degrees at most $\|D\|$; multiplication by c does not increase them. Theorem 6.1 therefore receives a rational function of that degree bound. If its output a has coefficient and denominator degrees at most E_a , the pole bound from Section 5 gives

$$\|\text{div}(a)\| \leq 4q(E_a + N).$$

Thus $\|E\| \leq \|D\| + 4q(E_a + N)$; the sum of the absolute degrees of the partial sums in (7.3) is at most $q^2\|E\|$. The maximum and final reduction consequently have polynomial size. Their ideal operations are precisely those of Proposition 5.1. Scalar pivots, valuations, and all other choices are determined by zero tests. If an instruction differs between root components, simultaneous execution returns the resulting factor; otherwise the entire construction proceeds over the product algebra. This proves the complexity and simultaneous-execution claims. $\square$

8 Splitting an odd number of roots

We now combine the geometry of Section 4 with the divisor algorithms. The input is a square-free polynomial already known to split into linear factors, but its roots are not given. Computation over its coordinate algebra treats all roots at once. If a scalar zero test behaves differently at two roots, it immediately produces a factor by Lemma 2.1. Proposition 4.6

supplies the reason that a distinction must occur. Our task here is to construct its input classes and perform its label tests within the promised bound.

Proposition 8.1 (Odd-degree splitting). *Let p be prime, $L = \lceil \log_2 p \rceil$, and $B = 20 + (n + 1)(L + 1)$. Suppose that $p > B^{200000}$ and that $F \in \mathbb{F}_p[X]$ is monic, square-free, totally split, and of odd degree $3 \leq N \leq n$. Choose an odd prime $q \mid N$. Suppose that the following data are supplied explicitly:*

- a field $K/\mathbb{F}_p$ of degree at most $q - 1$;
- a primitive q -th root of unity $\zeta \in K$;
- an element $\omega_K \in K^*$ of order $q^{v_q(|K|-1)}$.

There is a deterministic algorithm, denoted by OddSplit, that returns a nontrivial proper monic divisor of F over $\mathbb{F}_p$ in a number of bit operations bounded by a fixed polynomial in B .

8.1 The working field and the computational operations

Put

$$r = v_q(N), \quad m = 1 + (q - 1)r, \quad Q_K = |K|, \quad s_K = v_q(Q_K - 1). \quad (8.1)$$

Construct

$$k = K[T]/(T^{q^r} - \omega_K), \quad \omega = T \bmod (T^{q^r} - \omega_K). \quad (8.2)$$

This quotient is a field. Indeed, every root of the binomial has exact order q^{s_K+r} . For every positive integer b , the elementary lifting-the-exponent identity gives

$$v_q(Q_K^b - 1) = s_K + v_q(b).$$

Thus the smallest extension of K containing such a root has degree q^r . The binomial has that degree and is irreducible. The same identity shows that ω generates the entire q -primary subgroup of k^* . Notice the bounds

$$q^r \leq N, \quad m \leq q^r \leq N, \quad [k : \mathbb{F}_p] \leq N(q - 1). \quad (8.3)$$

The middle inequality follows from $q^r = (1 + (q - 1))^r \geq 1 + r(q - 1)$.

Use the curve C/K and its base extension to k from Section 4.1. Thus $Y^q = F(X)$, $P_i = (x_i, 0)$, $\lambda = 1 - \sigma$, and $e = (\sigma - 1)\infty_0$. Work simultaneously over

$$A = k[T_1]/F(T_1) \simeq \prod_{i=1}^N k.$$

The element $T_1 \bmod F$ provides the known ramification point in each component computation. The labels i and values x_i are used only in the correctness proof.

Here are the precise divisor operations that the procedure uses. The divisor arithmetic of Proposition 5.1 adds divisors, applies σ , and reduces a degree-zero divisor to an equivalent one of absolute degree at most $2g$. Lemma 5.3 sums a simultaneously represented family across the components of A , without finding those components. Lemma 5.4 finds a defining function for a principal divisor of the form $(1 - \sigma)D - W$, even when the infinity divisor W has large integer coefficients. It handles those coefficients by addition and doubling circuits and retains the function as a product circuit. Valuations and leading values at P_i can be computed from this circuit.

The procedure `LambdaDivide` of Proposition 7.1 takes a reduced degree-zero divisor D and a known ramification point P_i . Its promise is that $[D]$ has a λ -division represented by a divisor over k . It returns a reduced degree-zero divisor D' over k satisfying

$$\lambda[D'] = [D].$$

All these procedures use field arithmetic and scalar zero tests, so they admit the simultaneous execution just described. At any test that distinguishes components, `OddSplit` stops and returns the resulting factor.

8.2 The procedure

The algorithm has three stages: lift the ramification classes, sum the lifts, and test labels. We first give the complete instructions and then prove that the promise for every call holds and that a factor must be returned.

1. *Lift each ramification class.* In simultaneous execution over A , start with $D = P_i - \infty_0$ in component i . Apply `LambdaDivide`(D, P_i) successively $m - 1$ times, each time replacing D by its output. Write $d_{i,t}$ for the class after $t - 1$ divisions, so $d_{i,1} = [P_i - \infty_0]$.
2. *Sum the lifts and apply successive powers of λ .* Use the simultaneous divisor-sum operation to obtain a divisor representing

$$d_m = \sum_{i=1}^N d_{i,m}.$$

Reduce it, obtaining D_m . For $t = m - 1, m - 2, \dots, 1$, reduce $(1 - \sigma)D_{t+1}$ to obtain D_t . Thus, writing $d_t = [D_t]$,

$$d_t = \lambda^{m-t} d_m. \quad (8.4)$$

Compute the integral infinity divisors

$$W_t = \lambda^{-(t-1)} N e, \quad 1 \leq t \leq m. \quad (8.5)$$

Here the inverse notation denotes the unique solution in the lattice Λ , not division of divisor classes. These solutions exist because $q^r \mid N$ and q is an associate of λ^{q-1} . One may compute them successively by solving $(1 - \sigma)W_{t+1} = W_t$ on the cyclic infinity coordinates.

3. *Test the labels.* For $t = 1, 2, \dots, m$, perform the following operations. The relation $\lambda d_t = [W_t]$ will hold at every test that is reached. Find a function $h_t \in k(C)^*$ satisfying

$$\operatorname{div}(h_t) = (1 - \sigma)D_t - W_t. \quad (8.6)$$

It may be stored by the product-circuit method above. Evaluate $h_t(P_i)$ and $v_{P_i}(D_t)$ simultaneously. The value $h_t(P_i)^q$ is independent of i ; extract it as a scalar $c_t \in k^*$ from A . Use `UnitRoot` from Lemma 2.2 to find $\gamma_t \in k^*$ with $\gamma_t^q = c_t$. For each component, compare $h_t(P_i)/\gamma_t$ with $1, \zeta, \dots, \zeta^{q-1}$ and form

$$l_{i,t} = v_{P_i}(D_t) - \log_\zeta(h_t(P_i)/\gamma_t) \pmod{q}. \quad (8.7)$$

If these labels differ, return a nontrivial proper gcd with F corresponding to one of their values. If they all agree, proceed to the next test.

Every factor obtained by simultaneous execution lies in $\mathbb{F}_p[X]$. Indeed, although the gcd may be computed over k , it is the monic product of $X - x_i$ over a subset of roots $x_i \in \mathbb{F}_p$. Thus no additional descent of the output factor is required.

8.3 Correctness and cost

Proof of Proposition 8.1. Suppose that no earlier scalar test has returned a factor. The elements ζ and ω supply the primary-group data required by Theorem 6.1. In every division, its norm input is a constant multiple of the function j in Lemma 5.2. Since the input divisor has absolute degree at most $2g$, the reduced numerator and denominator of j have degrees at most $2g$. The norm solver's characteristic hypothesis therefore follows from $(N + 4g + 1)^2 < B^6 < p$. Corollary 4.5, with the r, m, k of (8.1)–(8.2), proves the promise for every invocation of LambdaDivide in the first stage. This remains true for the particular divisions chosen by the algorithm: every successive geometric division has a representative over k . We therefore have

$$\lambda^t d_{i,t} = [e] \quad (1 \leq t \leq m).$$

The simultaneous sum in the second stage also satisfies its scalar-grid requirement. A reduced divisor has absolute degree at most $2g$, so its compressed ideal bound has degree at most $2g$ and uses at most $b = 4qg + 1$ generators. Thus

$$N(b - 1) \leq 4Nqg = 2Nq(q - 1)(N - 2) < B^6 < p,$$

as required by Lemma 5.3. Summing the equalities $\lambda^m d_{i,m} = [e]$ gives

$$\lambda^m d_m = [Ne]. \tag{8.8}$$

Equation (8.8) is the hypothesis of Proposition 4.6. The classes in (8.4), their reduced representatives D_t , and the lattice divisors in (8.5) are exactly its inputs. That proposition proves that, at every reached test, the divisor in (8.6) is principal over k , the function is a unit at all P_i , and its norm is a q -th power in k^* . Lemma 5.4 computes h_t and its values, and Lemma 2.2 computes γ_t with its promise thus verified. The logarithms in (8.7) require only the stated q comparisons. These are precisely the labels of Proposition 4.6, which applies to all choices made by the deterministic routines. It guarantees a nonconstant label vector by test m , unless an earlier zero test has already returned a factor. Therefore the procedure returns a nontrivial proper factor.

For the running time, (8.3) bounds the extension degree and the number of divisions and tests by polynomials in n . All divisors D_t and all divisors retained between divisions are reduced, with absolute degree at most $2g$. The sole potentially large divisors are the W_t , and they have short integer descriptions. More explicitly, if $W = \sum_j a_j \infty_j$, solving $(1 - \sigma)W' = W$ amounts to taking partial sums of the a_j and subtracting their mean. Whenever the integral solution in Λ exists, this increases the maximum absolute coefficient by at most a factor $2q$. Hence every coefficient of every W_t has magnitude at most $N(2q)^m$, and thus bit length $O(\log N + m \log q)$. The principal-divisor computations use those coefficients through binary circuits, as guaranteed by Lemma 5.4. All remaining loops and field operations have polynomial cost; the explicit combined bit bound is derived in Section 10. This proves the claimed uniform polynomial bound. $\square$

9 From a splitting procedure to complete factorization

We now assemble the splitting procedures into an algorithm for an arbitrary nonzero polynomial over $\mathbf{F}_p$. The driver has two jobs: handle multiplicities without restrictions on the characteristic, and turn a reducible square-free polynomial into a totally split polynomial to which the earlier procedures apply. Its degree restrictions also explain why the primary table can be constructed in increasing order without circularity.

For a degree bound b , a *primary table through b* consists of a generator of the full 2-primary subgroup of $\mathbf{F}_p^*$, when $b \geq 2$, and, for each odd prime $q \leq b$, the field $K_q/\mathbf{F}_p$, the element ζ_q of order q , and the full q -primary generator ω_q constructed in Proposition 3.2. This is precisely the data consumed by the splitting procedures. All calls in this section are over $\mathbf{F}_p$, even when the splitting procedure itself uses an extension field.

9.1 Separating the factors through the Berlekamp algebra

Let $h \in \mathbf{F}_p[X]$ be monic and square-free, of degree $d \geq 1$. The Berlekamp algebra [2] in $R_h = \mathbf{F}_p[X]/h$ is

$$\mathcal{B}_h = \{a \in R_h : a^p = a\}. \quad (9.1)$$

The Frobenius map is $\mathbf{F}_p$ -linear, so a basis of $\mathcal{B}_h$ is computed by powering the standard basis of R_h and taking the kernel of Frobenius minus the identity. Products of basis elements are computed in R_h and expressed in this basis by linear algebra.

If $h = \prod_{i=1}^r h_i$ is its unknown irreducible factorization, the Chinese remainder theorem identifies

$$R_h \simeq \prod_{i=1}^r \mathbf{F}_p[X]/h_i, \quad \mathcal{B}_h \simeq \mathbf{F}_p^r.$$

Indeed the Frobenius-fixed elements of each finite field component are exactly $\mathbf{F}_p$. Thus $r = \dim \mathcal{B}_h$, and h is irreducible if and only if $r = 1$.

Lemma 9.1 (A separating element). *Let $h \in \mathbf{F}_p[X]$ be monic and square-free of degree d , let $r = \dim \mathcal{B}_h \geq 2$, and suppose $p > r^3$. From any ordered basis $b_0, \dots, b_{r-1}$ of $\mathcal{B}_h$, one can find an integer $t \in \{0, \dots, r^3\}$ such that the characteristic polynomial F_t of multiplication by*

$$b(t) = \sum_{j=0}^{r-1} t^j b_j \quad \text{on } \mathcal{B}_h$$

is square-free. The polynomial F_t is monic, totally split over $\mathbf{F}_p$, and has degree r . Every nontrivial proper monic divisor S of F_t gives a nontrivial proper factor $\gcd(h, S(b(t)))$ of h . The construction has bit complexity polynomial in d and $\log p$.

Proof. In $\mathcal{B}_h \simeq \mathbf{F}_p^r$, multiplication by $b(t)$ is diagonal with its r coordinates as diagonal entries. For any two coordinates their difference is a polynomial in t of degree at most $r - 1$. This polynomial is nonzero: otherwise the two coordinate functionals agree on every basis vector and hence on the whole algebra, contrary to the product description. There are therefore at most

$$\binom{r}{2}(r-1) < r^3$$

parameters at which any pair of coordinates coincide. Since $p > r^3$, the listed parameters are distinct elements of the field, and at least one has all coordinates different. This is exactly the condition that F_t be square-free, which is tested by $\gcd(F_t, F'_t) = 1$.

A proper factor S selects a nonempty proper subset of these distinct coordinates. The element $S(b(t)) \in R_h$ vanishes precisely on the corresponding irreducible components, so its gcd with h selects precisely the same nonempty proper subset of the factors h_i . Polynomial powering, linear algebra, determinants, and gcds of the displayed degrees implement all instructions within the stated bound. $\square$

9.2 The recursive driver

Fix the degree n of the original input, put

$$L = \lceil \log_2 p \rceil, \quad B = 20 + (n + 1)(L + 1), \quad (9.2)$$

and use the same B throughout preprocessing and recursion. Call the characteristic *small* when $p \leq B^{200000}$. For larger p , the inequalities needed for the separating-element search and all finite-field searches in the splitting procedures follow from the size bounds in Section 10.

The procedure $\text{Split}(F)$ receives a monic, square-free, totally split polynomial of degree $N \geq 2$ in the large-characteristic case. If N is even, use Proposition 2.3. If N is odd, use Proposition 8.1. In either case it returns a nontrivial proper monic divisor, and uses table entries only for primes at most N .

Here is the complete recursive procedure. Its input h is monic; its output is a list of distinct monic irreducible factors with multiplicities. Repeated occurrences of the same monic polynomial are merged by comparing their coefficient lists.

Procedure $\text{Factor}(h)$.

1. If $\deg h = 0$, return the empty list. If $\deg h = 1$, return h with multiplicity one.
2. Compute the formal derivative h' . If $h' = 0$, write $h = g^p$ by retaining the coefficients in positions divisible by p , recursively factor g , multiply every returned multiplicity by p , and return the resulting list.
3. If $h' \neq 0$, compute $d_0 = \gcd(h, h')$. When $d_0 \neq 1$, recursively factor d_0 and h/d_0 , merge the two lists by adding multiplicities of equal factors, and return the merged list.
4. Now h is square-free. Compute $\mathcal{B}_h$ and $r = \dim \mathcal{B}_h$. If $r = 1$, return h with multiplicity one.
5. If the characteristic is small and $r \geq 2$, choose a nonscalar $b \in \mathcal{B}_h$. Test $\gcd(h, b - a)$ for $a = 0, \dots, p - 1$ until a nontrivial proper gcd is found. Denote it by h_1 .
6. If the characteristic is large and $r \geq 2$, use Lemma 9.1 to obtain $b(t)$ and F_t . Compute $S = \text{Split}(F_t)$ and set $h_1 = \gcd(h, S(b(t)))$.
7. In either splitting case, recursively factor h_1 and h/h_1 , and return the union of the resulting lists.

A nonscalar element in Step 5 is found by testing the computed basis against the one-dimensional space $\mathbf{F}_p \cdot 1$. In Step 2 the coefficient list is merely scanned; no loop of length p is needed to form g .

Proposition 9.2 (Correctness and the table contract). *Fix n , p , and B as in Equation (9.2). For every monic $h \in \mathbf{F}_p[X]$ of degree $b \leq n$, $\text{Factor}(h)$ terminates and returns its complete irreducible factorization. In the large-characteristic case this conclusion assumes the primary table through b ; no entry for a prime larger than b is used. The number of recursive calls for a single input is $O((b + 1)^2)$.*

Proof. We argue by degree. The two base cases are immediate. In characteristic p , the derivative vanishes precisely when all nonconstant exponents are divisible by p . Every coefficient of $\mathbf{F}_p$ is its own p -th root, so the polynomial g in Step 2 satisfies $h = g^p$ exactly and has smaller degree. The rule for its multiplicities follows.

In Step 3 a nonconstant gcd is proper because $\deg h' < \deg h$. Both recursive arguments have smaller positive degree and their product is h . They need not be relatively prime; adding multiplicities handles exactly that possibility.

After these steps, h is square-free. Equation (9.1) and its product description prove the irreducibility test in Step 4. For Step 5, a nonscalar b has at least two different coordinates in $\mathbf{F}_p^r$. Taking a equal to one of its coordinate values makes $\gcd(h, b - a)$ nonconstant and proper, so the scalar search succeeds. Step 6 succeeds by Lemma 9.1 and the two splitting propositions. Both final recursive arguments again have smaller positive degree and product h .

Every call to a splitter has degree $r \leq \deg h$ and requires only primes at most that degree. Hence induction also proves the stated restriction on table entries. Along any recursion path degrees decrease strictly. At a binary branch the child degrees sum to the parent degree; at a unary branch the degree decreases. Consequently there are at most $b + 1$ levels and at most $O(b + 1)$ calls at each level, which gives the call bound. $\square$

9.3 Preprocessing and the original input

The table restriction in Proposition 9.2 closes the apparent dependency between factoring and constructing the table. Enumerate primes $q \leq n$ in increasing order by trial division. At prime q , carry out Proposition 3.2 with the supplied auxiliary prime ℓ_q . Its only factorization call, for odd q , is on the cyclotomic polynomial Φ_q of degree $q - 1$. Proposition 9.2 therefore uses only table entries already constructed. At $q = 2$ no factorization call is required. Induction over these primes constructs the whole table before the final call on the input polynomial. A recursive **Factor** call uses the available table and never restarts this preprocessing.

For the original nonzero polynomial f , first read its degree and leading coefficient c . A constant input returns $(c, \emptyset)$ immediately. For positive degree normalize to $h = c^{-1}f$. Degrees zero and one require no table. In the small-characteristic case use the scalar-search driver directly. Otherwise construct the table once as above and apply **Factor**(h). Its output (g_i, e_i) satisfies

$$f = c \prod_i g_i^{e_i},$$

with distinct monic irreducible g_i and positive integer e_i , exactly as required. Each exponent is at most n and is stored in binary.

Apart from the splitting calls, the driver consists of polynomial arithmetic and linear algebra of degree at most n , together with the displayed parameter searches. Scalar enumeration in the small case costs at most B^{200000} trials per call. In the large case the separating-element search has at most $n^3 + 1$ trials. Across the at most n table entries and the final factorization, Proposition 9.2 gives $O((n + 1)^3)$ recursive calls. Section 10 combines these bounds with the sizes of the norm, divisor, and table computations. Thus the complete algorithm has polynomial bit cost in n , L , and the largest supplied auxiliary prime; obtaining suitably small auxiliary primes is the separate arithmetic issue considered later.

10 Uniform bit complexity

We now charge the constructions to the dense binary input. The main point is that neither the degree of the curve nor the order of a primary subgroup is treated as a constant. A second issue is the large infinity divisor in the final label tests: its coefficients are stored in binary, and the corresponding functions are evaluated from circuits rather than expanded.

Throughout this section, $B = 20 + (n + 1)(L + 1)$ is the parameter in Equation (1.1). We use dense polynomial arithmetic and ordinary Gaussian elimination. We first count bit operations with indexed arrays, charging for binary addresses as well as arithmetic, and

then allow a quadratic overhead for a sequential bit implementation. The exponents below deliberately allow substantial slack; their purpose is to give one uniform bound.

Proposition 10.1. *Let E be the largest supplied auxiliary prime, or 2 when none is needed, as in Theorem 1.2. The complete algorithm uses*

$$O\left(B^{500000} + B^{100}E^{20}\right) \quad (10.1)$$

bit operations. The same bound applies if the auxiliary primes are found by upward search and E denotes the largest prime found. In particular, if $E = O(B^{20000000})$ with an absolute implied constant, the bit cost is $O(((n+1)L)^{10^{12}})$.

Proof. We first analyze the branch $p > B^{200000}$. Let $N \leq n$ be the degree of a polynomial passed to the splitting procedure. In an odd-degree split, the prime $q \mid N$, the integer $r = v_q(N)$, the extension field k , and the genus g satisfy

$$\begin{aligned} q, N, L &\leq B, & [k : \mathbf{F}_p] &\leq N(q-1) < B^2, \\ 2g &< B^2, & m = (q-1)r + 1 &< B^2. \end{aligned} \quad (10.2)$$

The intermediate fields used in making one table entry also have degree at most B^2 after the invariant algebra has been extracted.

Scalar arithmetic and rational functions. A scalar of k has fewer than B^3 bits in a basis over $\mathbf{F}_p$. A multiplication table, or the equivalent tower representation, gives addition, multiplication, and inversion by schoolbook arithmetic and linear algebra. The algebra $A = k[T]/F$ has dimension N over k ; its zero tests with split detection use polynomial gcds, as in Lemma 2.1. Allowing these tests and inversions, we may charge $O(B^{40})$ bit operations to each scalar instruction over k or over its simultaneous model A . This bound also covers scalar arithmetic in the even-degree algebra of dimension N^2 over $\mathbf{F}_p$. The auxiliary algebras of dimension comparable to ℓ_q are charged separately below.

In calculations over $k(X)$, fractions are reduced after arithmetic operations. More specifically, a linear system can first be given one common polynomial denominator. For a polynomial matrix of dimension at most d and entry degrees at most a , all its minors have degree at most da . Pivotal elimination entries are ratios of minors, and solutions may likewise be obtained by minors after selecting independent rows and columns. We use these degree bounds throughout; unreduced expression trees are never used as representations of rational functions. Polynomial determinants can be computed by evaluation and interpolation, with the grid sizes verified at the end of the proof.

One norm equation. Each call to NormSolve from Theorem 6.1 arises from a divisor of absolute degree at most $2g$, or from the initial divisor $P_i - \infty_0$. Compression expresses its pushforward as $j = h^+/h^-$, with numerator and denominator degrees at most the absolute degree of that divisor. Multiplication by the normalizing constant does not change degrees. Thus the parameter D_G , the maximum of the numerator and denominator degrees of $G = cj$, is at most B^3 .

Here is a size account for the linear construction of global sections of the order. The polynomial H is the product of the distinct finite bad places, and $M = 10q^2(N + D_G + 1)$ is the local valuation bound used in that construction.

Quantity	Bound
D_G	B^3
$\deg H \leq N + 2D_G$	B^4
M	B^7
Section numerator and common denominator degrees	B^{12}
Length of each truncated residue algebra	B^{13}
Number of rows and columns in the section system	B^{17}

For completeness, the section coordinates have the form P_{ab}/H^M , with $\deg P_{ab} \leq M(\deg H + 1)$. There are q^2 coordinates. The local conditions involve q^2 matrix entries at precision at most $3M + 3$; summing the residue degrees over all finite groups gives $\deg H$. Infinity contributes the same precision without a residue-degree factor. These observations give the last three rows of the table, including a bound on all scalar equations, rather than only on the number of groups of places.

The root computations in these residue algebras also have bounded sizes. Their square-free moduli have degree at most B^4 . In Lemma 2.2, every extension degree d is at most B^4 , so the bit length of $|k|^d$, and hence the number of primary digits, is at most B^8 . At each digit level there are at most B^4 nonempty pieces, each testing at most $q \leq B$ digits. The kernel-combination search uses at most B^9 scalars. Thus neither the numerical value of $|k|^d$ nor the order of its primary subgroup occurs as a loop length.

Let b be the global section lifting the selected rank-one matrix at infinity. To form the idempotent b^q , retain one common denominator in the standard cyclic-algebra basis. Multiplication by b contributes its denominator and, in reducing powers of the cyclic generator, at most one additional denominator of G . The numerator degrees grow by at most the corresponding numerator degrees and a further $O(qN)$ from reducing powers of Y . Repeating at most q times therefore keeps all degrees below B^{15} . The system $Ve' = he'$ for h has at most q unknown coefficients and q^2 equations. The ratio-of-minors bound gives a common denominator representation

$$h = \frac{1}{d_h(X)} \sum_{j=0}^{q-1} a_j(X) Y^j, \quad \deg a_j, \deg d_h \leq B^{20}.$$

The pole estimate for a function in this representation implies $\|\operatorname{div} h\| \leq B^{40}$, where $\|D\|$ denotes the absolute degree of a divisor.

These sizes also bound the number of operations. Put $U = B^{25}$. All polynomial lengths, system dimensions, truncation lengths, and integer bit lengths just listed are below U . A dense polynomial or truncated-series operation costs at most $O(U^4)$ scalar operations; ordinary linear algebra is cubic in its scalar dimensions. The loops for residue roots range over groups, degrees, digit levels, nonempty pieces, possible digits, and powering positions, each with at most U choices. Constructing local conditions ranges over groups, basis entries, ansatz coefficients, and precision positions. Even coefficient-by-coefficient Hensel lifting and separate construction of every matrix entry are covered by $O(U^{30})$ scalar and integer operations. Charging the scalar cost above, and schoolbook costs for the indicated integers, gives the convenient bound

$$\text{bit cost of one NormSolve call} = O(B^{2000}). \quad (10.3)$$

Divisor arithmetic and reduction. After a norm solution in `LambdaDivide`, the divisor $E_D = D - \operatorname{div} h$ has absolute degree at most B^{41} . Each orbit partial sum has at most q terms. The absolute degree of their coefficientwise maximum is bounded by the sum of their absolute degrees, so allowing the subtraction of $(\deg V)P_i$ leaves a reduction input of absolute degree at most B^{46} . All other reduction inputs satisfy the same bound: tracing adds at most N reduced divisors, and class additions and doublings reduce after every operation.

Compression gives a bound polynomial of degree at most $\|D\|$ for each stored divisor. The Riemann–Roch ansatz for reducing a divisor of size at most B^{46} consequently uses coefficient and denominator degrees at most B^{50} . The explicit local pole bounds give series precision at most B^{55} . Forming the associated principal ideal also requires an inverse

function. Its multiplication matrix has dimension q over $k(X)$, so the minor bound gives inverse coefficient degrees below B^{54} . Clearing denominators for the function and its inverse, and multiplying the bound polynomials of the ideals being combined, keeps temporary polynomial degrees below B^{60} . The subspace representation $H^{-1}O/HO$ then has dimension $2q \deg H < B^{65}$.

The ideal product used for tracing merits a separate check. Each component ideal has already been reduced, so its bound polynomial has degree at most B^3 and its generator list has length at most B^6 . The determinant formulas multiply at most N such functions. Before reduction by $Y^q = F(X)$, the degrees in each of X and Y are less than B^{10} . The number of scalar parameters in the generator search is at most $Nb + 1$, where b is the generator-list length. Thus tracing does not enumerate all choices of one generator from each ideal.

Put $U' = B^{200}$. The bounds just established place all lengths, dimensions, precisions, and intermediate generator lists below U' , including the lists of pairwise products in an ideal multiplication. For an ideal given by generators, closing their span under X and Y requires at most the ambient dimension many strict span enlargements. Each enlargement uses polynomial arithmetic and linear algebra. Intersections, inverse-ideal conditions, compression, and the Riemann–Roch conditions are linear systems in the same representations. Dense bivariate multiplication takes at most $O((U')^4)$ scalar operations; determinant interpolation uses at most two grids, each of length $O((U')^2)$. Together with the orbit sums and the trace construction, $O((U')^{30})$ scalar and integer operations bounds any one of these divisor procedures.

Large multiplicities and scalar grids. The infinity vectors W_t are the only divisors whose coefficient magnitudes need not be bounded by a fixed power of B . Their coefficients have magnitude at most $N(2q)^m$, so their binary lengths are at most B^4 , by Equation (10.2). The construction of $(1 - \sigma)D_t - W_t$ uses binary additions and doublings of divisor classes, reducing the representative after each instruction and retaining a circuit for the correcting function. Across all m label tests, at most B^{10} instructions suffice: there are at most q infinity differences per test and at most B^4 binary digits per coefficient. All leaf functions have the degree bounds for a single reduction established above.

At a ramification point, evaluate a leaf by its truncated expansion and propagate its valuation and leading coefficient through the circuit. Valuations are only added, doubled, or negated. Even allowing their magnitudes to double at every circuit instruction, their bit lengths are below B^{12} . Thus the circuit evaluation does not expand either a large function or a divisor with a large absolute degree. Charging $O(B^{100})$ for any of these integer operations is ample. Together with the bounds on m , the number of divisions, and the primitive procedures above, we obtain

$$\text{bit cost of one splitting call} = O(B^{20000}). \quad (10.4)$$

The even-degree splitter satisfies this bound as well: its algebra has dimension at most N^2 , its primary logarithms have at most L digits, and all exponentiations are binary.

Every scalar grid used above exists in the prime field. The separator search in the Berlekamp algebra uses at most $B^3 + 1$ scalars, the residue root search at most B^9 , and the trace search at most B^8 . For determinant interpolation, matrix dimensions and entry degrees are bounded by U' , so each grid needs at most $B^{400} + 1$ points. All these lengths are less than $p > B^{200000}$. The norm solver's assumption $p > (N + 2D_G + 1)^2$ follows from the same cutoff and $D_G \leq B^3$. In particular, its derivative-based square-free decompositions have polynomial degrees smaller than the characteristic.

The table and the factorization driver. For one auxiliary prime ℓ , the cyclotomic algebra has dimension $\ell - 1$. Its automorphisms are computed on the monomial basis, and their fixed

space can be found by stacking at most ℓ systems of at most ℓ equations each. Multiplying a basis of the fixed space and expressing its products in that basis produces the q -dimensional field representation used subsequently. Schoolbook polynomial arithmetic and rectangular elimination, without interpolation grids depending on ℓ , give the bound

$$O(\ell^{10} B^{40}) \tag{10.5}$$

for this construction, including the other field operations for the entry but excluding its recursive factorization call. Trial division verifies a supplied ℓ within the same bound. If instead it is found by upward search, trial division and the modular power test for every integer up to ℓ also fit Equation (10.5). Here residues modulo ℓ use $O(\log \ell)$ bits and the exponent test also uses the L -bit input p . Powers of $\log \ell$ are absorbed by the displayed power of ℓ ; no comparison between $\log \ell$ and $\log B$ is needed. There are at most B table entries, yielding $O(B^{41} E^{10})$ bit operations outside the recursive calls.

The table is built once, in increasing order of q . Its call to factor Φ_q has degree $q - 1$ and uses only entries already constructed. For a factorization input of degree at most n , every nontrivial recursive step decreases degree, and the child degrees in a binary split sum to the parent degree. Allowing $O(B^2)$ calls per input, and including all table inputs, $O(B^4)$ calls is more than sufficient. The Berlekamp linear systems, polynomial gcds, and evaluations preceding each split have polynomial dimensions at most those already charged. Equation (10.4) therefore bounds all this work by $O(B^{20004})$.

In the branch $p \leq B^{200000}$, no auxiliary table or curve is used. The extra scalar loop in the Berlekamp splitter has length at most p ; all polynomial, matrix, and recursion costs besides this loop are bounded, for example, by $O(B^{1000})$. Hence the entire small-prime branch takes $O(B^{201000})$ bit operations. The same elementary operations handle zero derivatives, multiplicities, constants, and normalization by the leading coefficient. Multiplicities never exceed n , so their integer bookkeeping is negligible in these bounds.

Combining the two branches gives

$$O(B^{201000} + B^{41} E^{10})$$

bit operations with indexed arrays. The total stored binary data is bounded by the same quantity; a sequential implementation can locate entries by scanning the stored data, incurring at most quadratic overhead. Squaring the preceding bound and enlarging the powers of B gives Equation (10.1). If $E \leq TB^{20000000}$ for a fixed absolute T , its second term is at most $T^{20} B^{400000100}$. Finally $B = O((n + 1)L)$, with an absolute constant, so the stated exponent 10^{12} suffices. $\square$

Proposition 10.1 completes the uniformity assertion in Theorem 1.2. It also identifies precisely where a number-theoretic input can enter the running time: only the numerical sizes of the auxiliary primes remain outside the polynomial bounds in B for the algebraic splitting construction.

11 Small auxiliary primes from a uniform zero-free strip

It remains to construct the auxiliary primes in (1.2) with numerical sizes polynomial in n and $\log p$. We use the following result from the companion paper. Its proof is not part of this paper; all dependence on that paper is isolated in this input.

Theorem 11.1 ([21], Theorem 1.2). *For every cyclotomic number field K containing the twelfth roots of unity, and every finite-order Hecke character χ of K , the Hecke L -function*

$L_K(s, \chi)$ has no zero in

$$\Re s > 1 - \delta, \quad \delta = 10^{-6}.$$

The principal character is included, with its pole at $s = 1$ permitted. There is no restriction on conductor or imaginary part.

The uniform width is what matters for the application: a zero-free region whose width deteriorates with the field discriminant or the height would not give the bound below by the same calculation.

11.1 A uniform smoothed prime-ideal estimate

Fix a nonzero nonnegative function $\varphi \in C_c^\infty((1, 2))$, and put

$$\Phi(s) = \int_0^\infty \varphi(t) t^{s-1} dt.$$

In particular $\Phi(1) > 0$. For a number field M of degree d_M and absolute discriminant D_M , define

$$\Psi_M(x) = \sum_{\mathfrak{p}} \sum_{j \geq 1} \log N \mathfrak{p} \varphi((N \mathfrak{p})^j / x),$$

where $\mathfrak{p}$ runs over nonzero prime ideals of its ring of integers.

Lemma 11.2. *If the Dedekind zeta function $\zeta_M(s)$ has no zero in $\Re s > 1 - \delta$, for the fixed δ in Theorem 11.1, then, uniformly for $x \geq 2$,*

$$\Psi_M(x) = x\Phi(1) + O_\varphi(x^{1-\delta}(\log D_M + d_M)). \quad (11.1)$$

The implied constant is independent of M .

Proof. We record the uniformity in the standard smoothed explicit formula; the same formula is used in [21, Section 9.2, Equation (9.9)]. Mellin inversion on $\Re s = 2$ expresses $\Psi_M(x)$ as the integral of $-(\zeta'_M/\zeta_M)(s)\Phi(s)x^s$. Moving the contour to $\Re s = -1/2$ gives

$$\Psi_M(x) = x\Phi(1) - \sum_{\rho} x^\rho \Phi(\rho) + O_\varphi(\log D_M + d_M),$$

where ρ ranges over the nontrivial zeros with multiplicities. The pole at 1 gives the first term. The possible zero at 0 has order at most d_M , and the integral on the new line is bounded using the functional equation and the absolutely convergent logarithmic derivative on $\Re s = 3/2$. These give the stated remainder.

For completeness, the uniform zero-counting estimate needed here is

$$\#\{\rho : j \leq |\Im \rho| < j+1\} \ll \log D_M + d_M \log(j+2), \quad j \geq 0.$$

An explicit zero-counting estimate with these uniform dependencies is given in [11, Corollary 1.2, Equation (1.3)]. For $j \geq 2$, subtracting its counts at heights $j-1$ and $j+2$ gives the displayed estimate; the count at height 3 handles $j < 2$. Its absolute implied constant does not depend on M . Integration by parts in the Mellin transform gives $|\Phi(\sigma+it)| \ll_\varphi (1+|t|)^{-3}$ uniformly for $0 \leq \sigma \leq 1$. Consequently

$$\sum_{\rho} |\Phi(\rho)| \ll_\varphi \log D_M + d_M.$$

The contour argument can be made first at heights avoiding zeros and then passed to the limit by this absolute convergence. The assumed strip places every nontrivial zero in $\Re \rho \leq 1 - \delta$. Multiplication by $x^{1-\delta}$ proves (11.1). $\square$

11.2 Detecting failure of complete splitting

Proposition 11.3. *There is an absolute constant c_0 such that, for every prime $p > B^{2000000}$ and every prime $q \leq n$, an auxiliary prime for (p, q) exists with*

$$\ell \leq c_0 B^{20000000}.$$

Proof. Set

$$K = \mathbf{Q}(\mu_{12q}), \quad M = K(p^{1/q}).$$

The large-characteristic assumption implies $p \nmid 12q$. Thus p is unramified in K , and $X^q - p$ is Eisenstein at any prime of K above p . It follows that $[M : K] = q$. Since $\mu_q \subset K$, the extension M/K is cyclic. Abelian zeta factorization [20, Chapter VII] writes ζ_M as the product of the q Hecke L -functions corresponding to its characters over K . Theorem 11.1 therefore supplies the strip required in Lemma 11.2 for both K and M .

We have $[K : \mathbf{Q}] \leq 12q$ and $[M : \mathbf{Q}] \leq 12q^2$. The cyclotomic discriminant formula gives $\log D_K \leq [K : \mathbf{Q}] \log(12q)$. The relative discriminant divides the power-basis discriminant ideal $(q^q p^{q-1})$, so

$$\log D_M \leq q \log D_K + [K : \mathbf{Q}](q \log q + (q-1) \log p).$$

Since $q, n, L \leq B$, both fields satisfy $\log D + [\cdot : \mathbf{Q}] \ll B^3$, with an absolute constant. By Lemma 11.2,

$$\Psi_K(x) - q^{-1} \Psi_M(x) = (1 - q^{-1})x\Phi(1) + O_\varphi(x^{1-\delta} B^3). \quad (11.2)$$

Take $x = TB^{20000000}$, where $T \geq 2$ is an absolute constant chosen large enough. Relative to the main term, the error is $O_\varphi(T^{-\delta} B^{-17})$. Hence the left side of (11.2) is at least $c_1 x$ for a fixed $c_1 > 0$.

We now bound all contributions that could occur without a prime of the required kind. On the K side, powers $j \geq 2$ and prime ideals of residue degree at least two contribute at most

$$O_\varphi([K : \mathbf{Q}] \sqrt{x} \log^2(2x)).$$

Indeed their underlying rational primes are at most $\sqrt{2x}$; the sum of residue degrees above each prime is at most $[K : \mathbf{Q}]$; and the number of relevant exponents is $O(\log(2x))$. The first-power degree-one prime ideals above $2, 3, p, q$ contribute $O_\varphi([K : \mathbf{Q}] \log(2x))$. Both bounds are $o(x)$ uniformly at the chosen scale, and enlarging the absolute T makes their sum less than $c_1 x$.

Every remaining first-power degree-one prime of K lies over a rational prime $\ell \equiv 1 \pmod{12q}$ outside $\{2, 3, p, q\}$. If $p^{(\ell-1)/q} \equiv 1 \pmod{\ell}$, the polynomial $X^q - p$ splits into distinct linear factors in its residue field $\mathbf{F}_\ell$. The prime ideal then splits completely in M/K , and its first-power contribution to Ψ_K is cancelled exactly by the q primes above it in $q^{-1} \Psi_M$. All remaining negative terms may be discarded when taking an upper bound for the difference. Thus, if no auxiliary prime lay in the support interval $(x, 2x)$, the preceding error contributions would bound that difference by less than $c_1 x$, a contradiction. We may take $c_0 = 2T$. $\square$

Proof of Theorem 1.1. For each prime $q \leq n$ in the large-characteristic branch, search upward for ℓ_q , testing primality by trial division and testing (1.2) by modular exponentiation. The search does not need the value of c_0 . Proposition 11.3 bounds its length. Construct the table and apply Theorem 1.2. The refined estimates in Section 10 bound the full sequential bit cost by $O(B^{400000100})$. Since $B = O((n+1)L)$, the displayed generous exponent follows. $\square$

The argument also identifies a consequence of the algebraic reduction alone. GRH for finite-order Hecke L -functions places their nontrivial zeros on $\Re s = 1/2$, and therefore implies the strip used above. Thus the reduction yields deterministic polynomial-time factorization under GRH independently of the companion’s proof of Theorem 11.1. Theorem 1.1 uses the cited theorem to supply the same input without that assumption.

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